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Unmanned vehicle recognition and threat management

Abstract

Systems and methods for automated unmanned aerial vehicle recognition. A multiplicity of receivers captures RF data and transmits the RF data to at least one node device. The at least one node device comprises a signal processing engine, a detection engine, a classification engine, and a direction finding engine. The at least one node device is configured with an artificial intelligence algorithm. The detection engine and classification engine are trained to detect and classify signals from unmanned vehicles and their controllers based on processed data from the signal processing engine. The direction finding engine is operable to provide lines of bearing for detected unmanned vehicles.

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continuation parent-doc US 16275575 20190214 US 10529241 20200107 child-doc US 16732811
continuation-in-part parent-doc US 18775710 20240717 US 12183213 20241231 child-doc US
18785936
continuation-in-part parent-doc US 18428606 20240131 US 12266272 20250401 child-doc US
18775710
continuation-in-part parent-doc US 16274933 20190213 US 10498951 20191203 child-doc US
16275575
continuation-in-part parent-doc US 16180690 20181105 US 10459020 20191029 child-doc US
16274933
continuation-in-part parent-doc US 15412982 20170123 US 10122479 20181106 child-doc US
16180690
us-provisional-application US 62722420 20180824
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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
4215345	12/1979	Robert et al.	N/A	N/A
4400700	12/1982	Rittenbach	N/A	N/A
4453137	12/1983	Rittenbach	N/A	N/A
4501020	12/1984	Wakeman	N/A	N/A
4581769	12/1985	Grimsley et al.	N/A	N/A
4638493	12/1986	Bishop et al.	N/A	N/A
4794325	12/1987	Britton et al.	N/A	N/A
4928106	12/1989	Ashjaee et al.	N/A	N/A
5103402	12/1991	Morton et al.	N/A	N/A
5134407	12/1991	Lorenz et al.	N/A	N/A
5144642	12/1991	Weinberg et al.	N/A	N/A
5166664	12/1991	Fish	N/A	N/A
5230087	12/1992	Meyer et al.	N/A	N/A

5293170	12/1993	Lorenz et al.	N/A	N/A
5343212	12/1993	Rose et al.	N/A	N/A
5393713	12/1994	Schwob	N/A	N/A
5416593	12/1994	Vercruysse	N/A	N/A
5448309	12/1994	Won	N/A	N/A
5506864	12/1995	Schilling	N/A	N/A
5513385	12/1995	Tanaka	N/A	N/A
5548809	12/1995	Lemson	N/A	N/A
5570099	12/1995	DesJardins	N/A	N/A
5589835	12/1995	Gildea et al.	N/A	N/A
5612703	12/1996	Mallinckrodt	N/A	N/A
5642732	12/1996	Wang	N/A	N/A
5831874	12/1997	Boone et al.	N/A	N/A
5835857	12/1997	Otten	N/A	N/A
5838906	12/1997	Doyle et al.	N/A	N/A
5846208	12/1997	Pichlmayr et al.	N/A	N/A
5856803	12/1998	Pevler	N/A	N/A
5936575	12/1998	Azzarelli et al.	N/A	N/A
6018312	12/1999	Haworth	N/A	N/A
6039692	12/1999	Kristoffersen	N/A	N/A
6085090	12/1999	Yee et al.	N/A	N/A
6115580	12/1999	Chuprun et al.	N/A	N/A
6134445	12/1999	Gould et al.	N/A	N/A
6144336	12/1999	Preston et al.	N/A	N/A
6157619	12/1999	Ozluturk et al.	N/A	N/A
6160511	12/1999	Pfeil et al.	N/A	N/A
6167277	12/1999	Kawamoto	N/A	N/A
6185309	12/2000	Attias	N/A	N/A
6188715	12/2000	Partyka	N/A	N/A
6191731	12/2000	McBurney et al.	N/A	N/A
6198414	12/2000	McPherson et al.	N/A	N/A
6249252	12/2000	Dupray	N/A	N/A
6286021	12/2000	Tran et al.	N/A	N/A
6296612	12/2000	Mo et al.	N/A	N/A
6304760	12/2000	Thomson et al.	N/A	N/A
6314366	12/2000	Farmakis et al.	N/A	N/A
6339396	12/2001	Mayersak	N/A	N/A
6384776	12/2001	Martin	N/A	N/A
6400647	12/2001	Huntress	N/A	N/A
6418131	12/2001	Snelling et al.	N/A	N/A
6430418	12/2001	Nivens et al.	N/A	N/A
6433671	12/2001	Nysen	N/A	N/A
6490318	12/2001	Larsson et al.	N/A	N/A
6492945	12/2001	Counselman, III et al.	N/A	N/A
6512788	12/2002	Kuhn et al.	N/A	N/A
6628231	12/2002	Mayersak	N/A	N/A
6677895	12/2003	Holt	N/A	N/A
6697439	12/2003	Trivedi et al.	N/A	N/A
6707910	12/2003	Valve et al.	N/A	N/A
6711404	12/2003	Arpee et al.	N/A	N/A

6741595	12/2003	Maher et al.	N/A	N/A
6771957	12/2003	Chitrapu	N/A	N/A
6785321	12/2003	Yang et al.	N/A	N/A
6850557	12/2004	Gronemeyer	N/A	N/A
6850735	12/2004	Sugar et al.	N/A	N/A
6859831	12/2004	Gelvin et al.	N/A	N/A
6861982	12/2004	Forstrom et al.	N/A	N/A
6876326	12/2004	Martorana	N/A	N/A
6898197	12/2004	Lavean	N/A	N/A
6898235	12/2004	Carlin et al.	N/A	N/A
6904269	12/2004	Deshpande et al.	N/A	N/A
6985437	12/2005	Vogel	N/A	N/A
6991514	12/2005	Meloni et al.	N/A	N/A
7035593	12/2005	Miller et al.	N/A	N/A
7043207	12/2005	Miyazaki	N/A	N/A
7049965	12/2005	Kelliher et al.	N/A	N/A
7110756	12/2005	Diener	N/A	N/A
7116943	12/2005	Sugar et al.	N/A	N/A
7146176	12/2005	Mchenry	N/A	N/A
7151790	12/2005	Patenaude et al.	N/A	N/A
7151938	12/2005	Weigand	N/A	N/A
7152025	12/2005	Lusky et al.	N/A	N/A
7162207	12/2006	Kursula et al.	N/A	N/A
7171161	12/2006	Miller	N/A	N/A
7187326	12/2006	Beadle et al.	N/A	N/A
7206350	12/2006	Korobkov et al.	N/A	N/A
7215716	12/2006	Smith	N/A	N/A
7254191	12/2006	Sugar et al.	N/A	N/A
7269151	12/2006	Diener et al.	N/A	N/A
7289733	12/2006	He	N/A	N/A
7292656	12/2006	Kloper et al.	N/A	N/A
7298327	12/2006	Dupray et al.	N/A	N/A
7340375	12/2007	Patenaud et al.	N/A	N/A
7366463	12/2007	Archer et al.	N/A	N/A
7408907	12/2007	Diener	N/A	N/A
7424268	12/2007	Diener et al.	N/A	N/A
7428270	12/2007	Dubuc et al.	N/A	N/A
7430254	12/2007	Anderson	N/A	N/A
7459898	12/2007	Woodings	N/A	N/A
7466960	12/2007	Sugar	N/A	N/A
7471683	12/2007	Maher, III et al.	N/A	N/A
7522917	12/2008	Purdy, Jr. et al.	N/A	N/A
7555262	12/2008	Brenner	N/A	N/A
7564816	12/2008	Mchenry et al.	N/A	N/A
7595754	12/2008	Mehta	N/A	N/A
7606335	12/2008	Kloper et al.	N/A	N/A
7606597	12/2008	Weigand	N/A	N/A
7620396	12/2008	Floam et al.	N/A	N/A
7653020	12/2009	Roberts	N/A	N/A
7676192	12/2009	Wilson	N/A	N/A

7692532	12/2009	Fischer et al.	N/A	N/A
7692573	12/2009	Funk	N/A	N/A
7702044	12/2009	Nallapureddy et al.	N/A	N/A
7725110	12/2009	Weigand	N/A	N/A
7728755	12/2009	Jocic	N/A	N/A
7801490	12/2009	Scherzer	N/A	N/A
7813742	12/2009	Mitchell	N/A	N/A
7835319	12/2009	Sugar	N/A	N/A
7865140	12/2010	Levien et al.	N/A	N/A
7893875	12/2010	Smith	N/A	N/A
7929508	12/2010	Yucek et al.	N/A	N/A
7933344	12/2010	Hassan et al.	N/A	N/A
7945215	12/2010	Tang	N/A	N/A
7953549	12/2010	Graham et al.	N/A	N/A
7965641	12/2010	Ben Letaief et al.	N/A	N/A
8001901	12/2010	Bass	N/A	N/A
8006195	12/2010	Woodings et al.	N/A	N/A
8023957	12/2010	Weigand	N/A	N/A
8026846	12/2010	Mcfadden et al.	N/A	N/A
8027249	12/2010	Mchenry et al.	N/A	N/A
8027690	12/2010	Shellhammer	N/A	N/A
8045654	12/2010	Anderson	N/A	N/A
8045660	12/2010	Gupta	N/A	N/A
8055204	12/2010	Livsics et al.	N/A	N/A
8059694	12/2010	Junell et al.	N/A	N/A
8060017	12/2010	Schlicht et al.	N/A	N/A
8060035	12/2010	Haykin	N/A	N/A
8060104	12/2010	Chaudhri et al.	N/A	N/A
8064840	12/2010	McHenry et al.	N/A	N/A
8077662	12/2010	Srinivasan et al.	N/A	N/A
RE43066	12/2011	McHenry	N/A	N/A
8094610	12/2011	Wang et al.	N/A	N/A
8107391	12/2011	Wu et al.	N/A	N/A
8125213	12/2011	Goguillon et al.	N/A	N/A
8131239	12/2011	Walker et al.	N/A	N/A
8134493	12/2011	Noble et al.	N/A	N/A
8151311	12/2011	Huffman et al.	N/A	N/A
8155039	12/2011	Wu et al.	N/A	N/A
8155649	12/2011	McHenry et al.	N/A	N/A
8160839	12/2011	Woodings et al.	N/A	N/A
8170577	12/2011	Singh	N/A	N/A
8175539	12/2011	Diener et al.	N/A	N/A
8184653	12/2011	Dain et al.	N/A	N/A
8193981	12/2011	Hwang et al.	N/A	N/A
8213868	12/2011	Du et al.	N/A	N/A
8224254	12/2011	Haykin	N/A	N/A
8229368	12/2011	Immendorf et al.	N/A	N/A
8233928	12/2011	Stanforth et al.	N/A	N/A
8238247	12/2011	Wu et al.	N/A	N/A
8249028	12/2011	Porras et al.	N/A	N/A

8249631	12/2011	Sawai	N/A	N/A
8260207	12/2011	Srinivasan et al.	N/A	N/A
8265684	12/2011	Sawai	N/A	N/A
8279786	12/2011	Smith et al.	N/A	N/A
8280433	12/2011	Quinn et al.	N/A	N/A
8289907	12/2011	Seidel et al.	N/A	N/A
8290503	12/2011	Sadek et al.	N/A	N/A
8295859	12/2011	Yarkan et al.	N/A	N/A
8295877	12/2011	Hui et al.	N/A	N/A
8301075	12/2011	Sherman et al.	N/A	N/A
8305215	12/2011	Markhovsky et al.	N/A	N/A
8311483	12/2011	Tillman et al.	N/A	N/A
8311509	12/2011	Feher	N/A	N/A
8315571	12/2011	Lindoff et al.	N/A	N/A
8320910	12/2011	Bobier	N/A	N/A
8326240	12/2011	Kadambe et al.	N/A	N/A
8326309	12/2011	Mody et al.	N/A	N/A
8326313	12/2011	McHenry et al.	N/A	N/A
8335204	12/2011	Samarasooriya et al.	N/A	N/A
8346273	12/2012	Weigand	N/A	N/A
8350970	12/2012	Birkett et al.	N/A	N/A
8352223	12/2012	Anthony et al.	N/A	N/A
8358723	12/2012	Hamkins et al.	N/A	N/A
8364188	12/2012	Srinivasan et al.	N/A	N/A
8369305	12/2012	Diener et al.	N/A	N/A
8373759	12/2012	Samarasooriya et al.	N/A	N/A
8391794	12/2012	Sawai et al.	N/A	N/A
8391796	12/2012	Srinivasan et al.	N/A	N/A
8401564	12/2012	Singh	N/A	N/A
8406776	12/2012	Jallon	N/A	N/A
8406780	12/2012	Mueck	N/A	N/A
RE44142	12/2012	Wilson	N/A	N/A
8421676	12/2012	Moshfeghi	N/A	N/A
8422453	12/2012	Abedi	N/A	N/A
8422958	12/2012	Du et al.	N/A	N/A
RE44237	12/2012	Mchenry	N/A	N/A
8437700	12/2012	Mody et al.	N/A	N/A
8442445	12/2012	Mody et al.	N/A	N/A
8447237	12/2012	Reial et al.	N/A	N/A
8451751	12/2012	Challapali et al.	N/A	N/A
8463195	12/2012	Shellhammer	N/A	N/A
8467353	12/2012	Proctor	N/A	N/A
8483155	12/2012	Banerjea et al.	N/A	N/A
8494464	12/2012	Kadambe et al.	N/A	N/A
8503955	12/2012	Kang et al.	N/A	N/A
8504087	12/2012	Stanforth et al.	N/A	N/A
8514729	12/2012	Blackwell	N/A	N/A
8515473	12/2012	Mody et al.	N/A	N/A
8520606	12/2012	Cleveland	N/A	N/A
RE44492	12/2012	Mchenry	N/A	N/A

8526974	12/2012	Olsson et al.	N/A	N/A
8532686	12/2012	Schmidt et al.	N/A	N/A
8538339	12/2012	Hu et al.	N/A	N/A
8548521	12/2012	Hui et al.	N/A	N/A
8554264	12/2012	Gibbons et al.	N/A	N/A
8559301	12/2012	Mchenry et al.	N/A	N/A
8565811	12/2012	Tan et al.	N/A	N/A
8599024	12/2012	Bloy	N/A	N/A
8718838	12/2013	Kokkeby et al.	N/A	N/A
8761051	12/2013	Brisebois et al.	N/A	N/A
8773966	12/2013	Petrovic et al.	N/A	N/A
8780968	12/2013	Garcia et al.	N/A	N/A
8798548	12/2013	Carbajal	N/A	N/A
8805291	12/2013	Garcia et al.	N/A	N/A
8818283	12/2013	McHenry et al.	N/A	N/A
8824536	12/2013	Garcia et al.	N/A	N/A
8843155	12/2013	Burton et al.	N/A	N/A
8886210	12/2013	Sugar	N/A	N/A
8941491	12/2014	Polk et al.	N/A	N/A
8977212	12/2014	Carbajal	N/A	N/A
9007262	12/2014	Witzgall	N/A	N/A
9008587	12/2014	Carbajal	N/A	N/A
9078162	12/2014	Garcia et al.	N/A	N/A
9143968	12/2014	Manku et al.	N/A	N/A
9185591	12/2014	Carbajal	N/A	N/A
9229102	12/2015	Wright et al.	N/A	N/A
9245378	12/2015	Villagomez et al.	N/A	N/A
9288683	12/2015	Garcia et al.	N/A	N/A
9356727	12/2015	Immendorf et al.	N/A	N/A
9397619	12/2015	Lozhkin	N/A	N/A
9412278	12/2015	Gong et al.	N/A	N/A
9413574	12/2015	Timofeev et al.	N/A	N/A
9414237	12/2015	Garcia et al.	N/A	N/A
9439078	12/2015	Menon et al.	N/A	N/A
9466881	12/2015	Berry et al.	N/A	N/A
9529360	12/2015	Melamed et al.	N/A	N/A
9537586	12/2016	Carbajal	N/A	N/A
9538040	12/2016	Goergen et al.	N/A	N/A
9572055	12/2016	Immendorf et al.	N/A	N/A
9635669	12/2016	Gormley et al.	N/A	N/A
9658341	12/2016	Mathews et al.	N/A	N/A
9674684	12/2016	Mendelson	N/A	N/A
9674836	12/2016	Gormley et al.	N/A	N/A
9686789	12/2016	Gormley et al.	N/A	N/A
9715009	12/2016	Parker et al.	N/A	N/A
9749069	12/2016	Garcia et al.	N/A	N/A
9755972	12/2016	Mao et al.	N/A	N/A
9767699	12/2016	Borghese et al.	N/A	N/A
9769834	12/2016	Immendorf et al.	N/A	N/A
9805273	12/2016	Seeber et al.	N/A	N/A

9819441	12/2016	Immendorf et al.	N/A	N/A
9858947	12/2017	Hearing et al.	N/A	N/A
9862489	12/2017	Weinstein et al.	N/A	N/A
9923700	12/2017	Gormley et al.	N/A	N/A
9942775	12/2017	Yun et al.	N/A	N/A
9973278	12/2017	Wang et al.	N/A	N/A
9989633	12/2017	Pandey et al.	N/A	N/A
9998243	12/2017	Garcia et al.	N/A	N/A
10027429	12/2017	Kiannejad	N/A	N/A
10104559	12/2017	Immendorf et al.	N/A	N/A
10157548	12/2017	Priest	N/A	N/A
10194324	12/2018	Yun et al.	N/A	N/A
10198955	12/2018	Boyd et al.	N/A	N/A
10227429	12/2018	Watanabe et al.	N/A	N/A
10235523	12/2018	Keller, III et al.	N/A	N/A
10241140	12/2018	Moinuddin	N/A	N/A
10251242	12/2018	Rosen et al.	N/A	N/A
10281570	12/2018	Parker et al.	N/A	N/A
10389616	12/2018	Ryan et al.	N/A	N/A
10393784	12/2018	Logan et al.	N/A	N/A
10408936	12/2018	Van Voorst	N/A	N/A
10459020	12/2018	Dzierwa et al.	N/A	N/A
10540905	12/2019	Bohanan et al.	N/A	N/A
10552738	12/2019	Holt et al.	N/A	N/A
10587352	12/2019	Kiannejad	N/A	N/A
10594034	12/2019	Tran et al.	N/A	N/A
10613209	12/2019	Emami et al.	N/A	N/A
10642813	12/2019	Lazier et al.	N/A	N/A
10698076	12/2019	Jones et al.	N/A	N/A
10700721	12/2019	Ayala et al.	N/A	N/A
10701574	12/2019	Gormley et al.	N/A	N/A
10764718	12/2019	Boettcher et al.	N/A	N/A
10784974	12/2019	Menon	N/A	N/A
10811771	12/2019	Tran et al.	N/A	N/A
10907940	12/2020	Parker et al.	N/A	N/A
10916845	12/2020	Tran et al.	N/A	N/A
10917797	12/2020	Menon et al.	N/A	N/A
11012340	12/2020	Ryan et al.	N/A	N/A
11035929	12/2020	Parker et al.	N/A	N/A
11063653	12/2020	Ottersten et al.	N/A	N/A
11190233	12/2020	Lo et al.	N/A	N/A
11223431	12/2021	Garcia et al.	N/A	N/A
11265652	12/2021	Kallai et al.	N/A	N/A
11321282	12/2021	Tran	N/A	N/A
11334807	12/2021	O'Shea et al.	N/A	N/A
11336011	12/2021	Tran et al.	N/A	N/A
11516071	12/2021	Karapantelakis et al.	N/A	N/A
11637641	12/2022	Garcia et al.	N/A	N/A
11663992	12/2022	Canberk et al.	N/A	N/A
11671839	12/2022	Guo et al.	N/A	N/A

11700304	12/2022	Brown, Jr. et al.	N/A	N/A
11757185	12/2022	Tran et al.	N/A	N/A
11777783	12/2022	Meirosu et al.	N/A	N/A
11791913	12/2022	Garcia et al.	N/A	N/A
11871103	12/2023	Kleinbeck	N/A	N/A
11880888	12/2023	Gold et al.	N/A	N/A
11889351	12/2023	Tagg	N/A	N/A
11910305	12/2023	Buyukdura	N/A	N/A
12095518	12/2023	Garcia et al.	N/A	N/A
12156037	12/2023	Montalvo	N/A	N/A
12183213	12/2023	Kleinbeck et al.	N/A	N/A
12184963	12/2023	Kleinbeck	N/A	N/A
12243431	12/2024	Kleinbeck et al.	N/A	N/A
2001/0000959	12/2000	Campana et al.	N/A	N/A
2001/0005423	12/2000	Rhoads	N/A	N/A
2001/0016503	12/2000	Kang	N/A	N/A
2001/0020220	12/2000	Kurosawa	N/A	N/A
2002/0044082	12/2001	Woodington et al.	N/A	N/A
2002/0070889	12/2001	Griffin et al.	N/A	N/A
2002/0072331	12/2001	Fischer et al.	N/A	N/A
2002/0097184	12/2001	Mayersak	N/A	N/A
2002/0119754	12/2001	Wakutsu et al.	N/A	N/A
2002/0161775	12/2001	Lasensky et al.	N/A	N/A
2002/0173341	12/2001	Abdelmonem et al.	N/A	N/A
2003/0013454	12/2002	Hunzinger	N/A	N/A
2003/0040277	12/2002	Deats	N/A	N/A
2003/0083091	12/2002	Nuutinen et al.	N/A	N/A
2003/0087648	12/2002	Mezhvinsky et al.	N/A	N/A
2003/0104831	12/2002	Razavilar et al.	N/A	N/A
2003/0144601	12/2002	Prichep	N/A	N/A
2003/0145008	12/2002	Burrell	N/A	N/A
2003/0145328	12/2002	Rabinowitz et al.	N/A	N/A
2003/0198304	12/2002	Sugar et al.	N/A	N/A
2003/0206640	12/2002	Malvar et al.	N/A	N/A
2003/0224801	12/2002	Lovberg et al.	N/A	N/A
2003/0232612	12/2002	Richards et al.	N/A	N/A
2004/0001688	12/2003	Shen	N/A	N/A
2004/0023674	12/2003	Miller	N/A	N/A
2004/0127214	12/2003	Reddy et al.	N/A	N/A
2004/0137915	12/2003	Diener et al.	N/A	N/A
2004/0147254	12/2003	Reddy et al.	N/A	N/A
2004/0171390	12/2003	Chitrapu	N/A	N/A
2004/0203725	12/2003	Lahav et al.	N/A	N/A
2004/0203826	12/2003	Sugar et al.	N/A	N/A
2004/0208238	12/2003	Thomas et al.	N/A	N/A
2004/0219885	12/2003	Sugar et al.	N/A	N/A
2004/0233100	12/2003	Dibble et al.	N/A	N/A
2005/0003828	12/2004	Sugar et al.	N/A	N/A
2005/0096026	12/2004	Chitrapu et al.	N/A	N/A
2005/0107102	12/2004	Yoon et al.	N/A	N/A

2005/0114023	12/2004	Williamson et al.	N/A	N/A
2005/0152317	12/2004	Awater et al.	N/A	N/A
2005/0159928	12/2004	Moser	N/A	N/A
2005/0176401	12/2004	Nanda et al.	N/A	N/A
2005/0185618	12/2004	Friday et al.	N/A	N/A
2005/0192727	12/2004	Shostak et al.	N/A	N/A
2005/0227625	12/2004	Diener	N/A	N/A
2005/0285792	12/2004	Sugar et al.	N/A	N/A
2006/0025118	12/2005	Chitrapu et al.	N/A	N/A
2006/0047704	12/2005	Gopalakrishnan	N/A	N/A
2006/0074558	12/2005	Williamson et al.	N/A	N/A
2006/0080040	12/2005	Garczarek et al.	N/A	N/A
2006/0111899	12/2005	Padhi et al.	N/A	N/A
2006/0128311	12/2005	Tesfai	N/A	N/A
2006/0133263	12/2005	Bernard et al.	N/A	N/A
2006/0199546	12/2005	Durgin	N/A	N/A
2006/0235574	12/2005	Lapinski et al.	N/A	N/A
2006/0238417	12/2005	Jendbro et al.	N/A	N/A
2006/0258347	12/2005	Chitrapu	N/A	N/A
2007/0003059	12/2006	Langelaar	380/241	G06T 1/0028
2007/0016412	12/2006	Mehrotra et al.	N/A	N/A
2007/0041481	12/2006	Malkemes et al.	N/A	N/A
2007/0049823	12/2006	Li	N/A	N/A
2007/0076657	12/2006	Woodings et al.	N/A	N/A
2007/0098089	12/2006	Li et al.	N/A	N/A
2007/0111746	12/2006	Anderson	N/A	N/A
2007/0149216	12/2006	Misikangas	N/A	N/A
2007/0168580	12/2006	Schumacher	N/A	N/A
2007/0171889	12/2006	Kwon et al.	N/A	N/A
2007/0203645	12/2006	Dees et al.	N/A	N/A
2007/0223419	12/2006	Ji et al.	N/A	N/A
2007/0233336	12/2006	Serguei et al.	N/A	N/A
2007/0233409	12/2006	Boyan et al.	N/A	N/A
2007/0273581	12/2006	Garrison et al.	N/A	N/A
2007/0293171	12/2006	Li et al.	N/A	N/A
2007/0296591	12/2006	Frederick et al.	N/A	N/A
2007/0297541	12/2006	Mcgehee	N/A	N/A
2008/0001735	12/2007	Tran	N/A	N/A
2008/0010040	12/2007	Mcgehee	N/A	N/A
2008/0090563	12/2007	Chitrapu	N/A	N/A
2008/0113634	12/2007	Gates et al.	N/A	N/A
2008/0123731	12/2007	Wegener	N/A	N/A
2008/0129367	12/2007	Murata et al.	N/A	N/A
2008/0130519	12/2007	Bahl et al.	N/A	N/A
2008/0133190	12/2007	Peretz et al.	N/A	N/A
2008/0180325	12/2007	Chung et al.	N/A	N/A
2008/0186235	12/2007	Struckman et al.	N/A	N/A
2008/0195584	12/2007	Nath et al.	N/A	N/A
2008/0209117	12/2007	Kajigaya	N/A	N/A

2008/0211481	12/2007	Chen	N/A	N/A
2008/0214903	12/2007	Orbach	N/A	N/A
2008/0252516	12/2007	Ho et al.	N/A	N/A
2008/0261509	12/2007	Sen	N/A	N/A
2008/0293353	12/2007	Mody et al.	N/A	N/A
2009/0006103	12/2008	Koishida et al.	N/A	N/A
2009/0011713	12/2008	Abusubaih et al.	N/A	N/A
2009/0018422	12/2008	Banet et al.	N/A	N/A
2009/0021420	12/2008	Sahinoglu	N/A	N/A
2009/0046003	12/2008	Tung et al.	N/A	N/A
2009/0046625	12/2008	Diener et al.	N/A	N/A
2009/0066578	12/2008	Beadle et al.	N/A	N/A
2009/0086993	12/2008	Kawaguchi et al.	N/A	N/A
2009/0103094	12/2008	Hilfiker et al.	N/A	N/A
2009/0111463	12/2008	Simms et al.	N/A	N/A
2009/0131067	12/2008	Aaron	N/A	N/A
2009/0135046	12/2008	Steele et al.	N/A	N/A
2009/0136052	12/2008	Hohlfeld et al.	N/A	N/A
2009/0143019	12/2008	Shellhammer	N/A	N/A
2009/0146881	12/2008	Mesecher	N/A	N/A
2009/0149202	12/2008	Hill et al.	N/A	N/A
2009/0190511	12/2008	Li et al.	N/A	N/A
2009/0207950	12/2008	Tsuruta et al.	N/A	N/A
2009/0224957	12/2008	Chung et al.	N/A	N/A
2009/0245327	12/2008	Michaels	N/A	N/A
2009/0278733	12/2008	Haworth	N/A	N/A
2009/0280748	12/2008	Shan et al.	N/A	N/A
2009/0282130	12/2008	Antoniou et al.	N/A	N/A
2009/0285173	12/2008	Koorapaty et al.	N/A	N/A
2009/0286563	12/2008	Ji et al.	N/A	N/A
2009/0322510	12/2008	Berger et al.	N/A	N/A
2010/0020707	12/2009	Woodings	N/A	N/A
2010/0044122	12/2009	Sleeman et al.	N/A	N/A
2010/0056200	12/2009	Tolonen	N/A	N/A
2010/0075704	12/2009	Mchenry et al.	N/A	N/A
2010/0109936	12/2009	Levy	N/A	N/A
2010/0142454	12/2009	Chang	N/A	N/A
2010/0150122	12/2009	Berger et al.	N/A	N/A
2010/0172443	12/2009	Shim et al.	N/A	N/A
2010/0173586	12/2009	Mchenry et al.	N/A	N/A
2010/0176988	12/2009	Maezawa et al.	N/A	N/A
2010/0177710	12/2009	Gutkin et al.	N/A	N/A
2010/0220011	12/2009	Heuser	N/A	N/A
2010/0253512	12/2009	Wagner et al.	N/A	N/A
2010/0255794	12/2009	Agnew	N/A	N/A
2010/0255801	12/2009	Gunasekara et al.	N/A	N/A
2010/0259998	12/2009	Kwon et al.	N/A	N/A
2010/0279680	12/2009	Reudink	N/A	N/A
2010/0292930	12/2009	Koster et al.	N/A	N/A
2010/0306249	12/2009	Hill et al.	N/A	N/A

2010/0309317	12/2009	Wu et al.	N/A	N/A
2010/0325621	12/2009	Andrade et al.	N/A	N/A
2011/0022342	12/2010	Pandharipande et al.	N/A	N/A
2011/0045781	12/2010	Shellhammer et al.	N/A	N/A
2011/0053604	12/2010	Kim et al.	N/A	N/A
2011/0059747	12/2010	Lindoff et al.	N/A	N/A
2011/0070885	12/2010	Ruuska et al.	N/A	N/A
2011/0074631	12/2010	Parker	N/A	N/A
2011/0077017	12/2010	Yu et al.	N/A	N/A
2011/0087639	12/2010	Gurney	N/A	N/A
2011/0090939	12/2010	Diener et al.	N/A	N/A
2011/0096770	12/2010	Henry	N/A	N/A
2011/0102258	12/2010	Underbrink et al.	N/A	N/A
2011/0111751	12/2010	Markhovsky et al.	N/A	N/A
2011/0116484	12/2010	Henry	N/A	N/A
2011/0117869	12/2010	Woodings	N/A	N/A
2011/0122855	12/2010	Henry	N/A	N/A
2011/0129006	12/2010	Jung et al.	N/A	N/A
2011/0131260	12/2010	Mody	N/A	N/A
2011/0134878	12/2010	Geiger et al.	N/A	N/A
2011/0136478	12/2010	Trigui	N/A	N/A
2011/0151876	12/2010	Ishii et al.	N/A	N/A
2011/0183621	12/2010	Quan et al.	N/A	N/A
2011/0183685	12/2010	Burton et al.	N/A	N/A
2011/0185059	12/2010	Adnani et al.	N/A	N/A
2011/0188544	12/2010	Ponnuswamy	N/A	N/A
2011/0235728	12/2010	Karabinis	N/A	N/A
2011/0237243	12/2010	Guvenc et al.	N/A	N/A
2011/0241923	12/2010	Chernukhin	N/A	N/A
2011/0273328	12/2010	Parker	N/A	N/A
2011/0286555	12/2010	Cho et al.	N/A	N/A
2011/0286604	12/2010	Matsuo	N/A	N/A
2011/0287779	12/2010	Harper	N/A	N/A
2011/0299481	12/2010	Kim et al.	N/A	N/A
2011/0300849	12/2010	Chan	N/A	N/A
2011/0319120	12/2010	Chen et al.	N/A	N/A
2012/0014332	12/2011	Smith et al.	N/A	N/A
2012/0032854	12/2011	Bull et al.	N/A	N/A
2012/0039284	12/2011	Barbieri et al.	N/A	N/A
2012/0040602	12/2011	Charland	N/A	N/A
2012/0047544	12/2011	Bouchard	N/A	N/A
2012/0052869	12/2011	Lindoff et al.	N/A	N/A
2012/0058775	12/2011	Dupray et al.	N/A	N/A
2012/0063302	12/2011	Damnjanovic et al.	N/A	N/A
2012/0071188	12/2011	Wang et al.	N/A	N/A
2012/0072986	12/2011	Livsics et al.	N/A	N/A
2012/0077510	12/2011	Chen et al.	N/A	N/A
2012/0078071	12/2011	Bohm et al.	N/A	N/A
2012/0081248	12/2011	Kennedy et al.	N/A	N/A
2012/0094681	12/2011	Freda et al.	N/A	N/A

2012/0100810	12/2011	Oksanen et al.	N/A	N/A
2012/0105066	12/2011	Marvin et al.	N/A	N/A
2012/0115522	12/2011	Nama et al.	N/A	N/A
2012/0115525	12/2011	Kang et al.	N/A	N/A
2012/0120892	12/2011	Freda et al.	N/A	N/A
2012/0129522	12/2011	Kim et al.	N/A	N/A
2012/0140236	12/2011	Babbitt et al.	N/A	N/A
2012/0142386	12/2011	Mody et al.	N/A	N/A
2012/0148068	12/2011	Chandra et al.	N/A	N/A
2012/0148069	12/2011	Bai et al.	N/A	N/A
2012/0155217	12/2011	Dellinger et al.	N/A	N/A
2012/0169424	12/2011	Pinarello et al.	N/A	N/A
2012/0182430	12/2011	Birkett et al.	N/A	N/A
2012/0195269	12/2011	Kang et al.	N/A	N/A
2012/0212628	12/2011	Wu et al.	N/A	N/A
2012/0214511	12/2011	Vartanian et al.	N/A	N/A
2012/0230214	12/2011	Kozisek et al.	N/A	N/A
2012/0246392	12/2011	Cheon	N/A	N/A
2012/0264388	12/2011	Guo et al.	N/A	N/A
2012/0264445	12/2011	Lee et al.	N/A	N/A
2012/0275354	12/2011	Villain	N/A	N/A
2012/0281000	12/2011	Woodings	N/A	N/A
2012/0282942	12/2011	Uusitalo et al.	N/A	N/A
2012/0295575	12/2011	Nam	N/A	N/A
2012/0302190	12/2011	Mchenry	N/A	N/A
2012/0302263	12/2011	Tinnakornsrisuphap et al.	N/A	N/A
2012/0309288	12/2011	Lu	N/A	N/A
2012/0321024	12/2011	Wasiewicz et al.	N/A	N/A
2012/0322487	12/2011	Stanforth	N/A	N/A
2013/0005240	12/2012	Novak et al.	N/A	N/A
2013/0005374	12/2012	Uusitalo et al.	N/A	N/A
2013/0012134	12/2012	Jin et al.	N/A	N/A
2013/0017794	12/2012	Kloper et al.	N/A	N/A
2013/0023285	12/2012	Markhovsky et al.	N/A	N/A
2013/0028111	12/2012	Dain et al.	N/A	N/A
2013/0035108	12/2012	Joslyn et al.	N/A	N/A
2013/0035128	12/2012	Chan et al.	N/A	N/A
2013/0045754	12/2012	Markhovsky et al.	N/A	N/A
2013/0052939	12/2012	Anniballi et al.	N/A	N/A
2013/0053054	12/2012	Lovitt et al.	N/A	N/A
2013/0062334	12/2012	Bilchinsky et al.	N/A	N/A
2013/0064197	12/2012	Novak et al.	N/A	N/A
2013/0064328	12/2012	Adnani et al.	N/A	N/A
2013/0070639	12/2012	Demura et al.	N/A	N/A
2013/0090071	12/2012	Abraham et al.	N/A	N/A
2013/0095843	12/2012	Smith et al.	N/A	N/A
2013/0100154	12/2012	Woodings et al.	N/A	N/A
2013/0103684	12/2012	Yee et al.	N/A	N/A
2013/0113659	12/2012	Morgan	N/A	N/A

2013/0165051	12/2012	Li et al.	N/A	N/A
2013/0165134	12/2012	Touag et al.	N/A	N/A
2013/0165170	12/2012	Kang	N/A	N/A
2013/0183989	12/2012	Hasegawa et al.	N/A	N/A
2013/0183994	12/2012	Ringstroem et al.	N/A	N/A
2013/0184022	12/2012	Schmidt	N/A	N/A
2013/0190003	12/2012	Smith et al.	N/A	N/A
2013/0190028	12/2012	Wang et al.	N/A	N/A
2013/0196677	12/2012	Smith et al.	N/A	N/A
2013/0208587	12/2012	Bala et al.	N/A	N/A
2013/0210457	12/2012	Kummetz	N/A	N/A
2013/0210473	12/2012	Weigand	N/A	N/A
2013/0217406	12/2012	Villardi et al.	N/A	N/A
2013/0217408	12/2012	Difazio et al.	N/A	N/A
2013/0217450	12/2012	Kanj et al.	N/A	N/A
2013/0231121	12/2012	Kwak et al.	N/A	N/A
2013/0237212	12/2012	Khayrallah et al.	N/A	N/A
2013/0242792	12/2012	Woodings	N/A	N/A
2013/0242934	12/2012	Ueda et al.	N/A	N/A
2013/0260703	12/2012	Actis et al.	N/A	N/A
2013/0265198	12/2012	Stroud	N/A	N/A
2013/0272436	12/2012	Makhlouf et al.	N/A	N/A
2013/0279556	12/2012	Seller	N/A	N/A
2013/0288734	12/2012	Mody et al.	N/A	N/A
2013/0309975	12/2012	Kpodzo et al.	N/A	N/A
2013/0315112	12/2012	Gormley et al.	N/A	N/A
2013/0329690	12/2012	Kim et al.	N/A	N/A
2013/0331114	12/2012	Gormley et al.	N/A	N/A
2014/0003547	12/2013	Williams et al.	N/A	N/A
2014/0015796	12/2013	Philipp	N/A	N/A
2014/0018683	12/2013	Park et al.	N/A	N/A
2014/0024405	12/2013	Qiu	N/A	N/A
2014/0064723	12/2013	Adles et al.	N/A	N/A
2014/0066060	12/2013	Ngai	N/A	N/A
2014/0073261	12/2013	Hassan et al.	N/A	N/A
2014/0086212	12/2013	Kafle et al.	N/A	N/A
2014/0128032	12/2013	Muthukumar	N/A	N/A
2014/0139374	12/2013	Wellman et al.	N/A	N/A
2014/0163309	12/2013	Bernhard et al.	N/A	N/A
2014/0199993	12/2013	Dhanda et al.	N/A	N/A
2014/0201367	12/2013	Trummer et al.	N/A	N/A
2014/0204766	12/2013	Immendorf et al.	N/A	N/A
2014/0206279	12/2013	Immendorf et al.	N/A	N/A
2014/0206307	12/2013	Maurer et al.	N/A	N/A
2014/0206343	12/2013	Immendorf et al.	N/A	N/A
2014/0207414	12/2013	Bordow	N/A	N/A
2014/0225590	12/2013	Jacobs	N/A	N/A
2014/0256268	12/2013	Olgaard	N/A	N/A
2014/0256370	12/2013	Gautier et al.	N/A	N/A
2014/0269374	12/2013	Abdelmonem et al.	N/A	N/A

2014/0269376	12/2013	Garcia et al.	N/A	N/A
2014/0274103	12/2013	Steer et al.	N/A	N/A
2014/0287100	12/2013	Libman	N/A	N/A
2014/0301216	12/2013	Immendorf et al.	N/A	N/A
2014/0302796	12/2013	Gormley et al.	N/A	N/A
2014/0335879	12/2013	Immendorf et al.	N/A	N/A
2014/0340684	12/2013	Edler et al.	N/A	N/A
2014/0342675	12/2013	Massarella et al.	N/A	N/A
2014/0348004	12/2013	Ponnuswamy	N/A	N/A
2014/0362934	12/2013	Kumar	N/A	N/A
2015/0016429	12/2014	Menon et al.	N/A	N/A
2015/0023329	12/2014	Jiang et al.	N/A	N/A
2015/0068296	12/2014	Lanza di Scalea	N/A	N/A
2015/0072633	12/2014	Massarella et al.	N/A	N/A
2015/0126181	12/2014	Breuer et al.	N/A	N/A
2015/0133058	12/2014	Livis et al.	N/A	N/A
2015/0150753	12/2014	Racette	N/A	N/A
2015/0156827	12/2014	Ibragimov et al.	N/A	N/A
2015/0170145	12/2014	Patel et al.	N/A	N/A
2015/0201385	12/2014	Mercer et al.	N/A	N/A
2015/0208312	12/2014	Fudaba	N/A	N/A
2015/0215794	12/2014	Gormley et al.	N/A	N/A
2015/0215949	12/2014	Gormley et al.	N/A	N/A
2015/0242384	12/2014	Reiter	N/A	N/A
2015/0248047	12/2014	Chakraborty	N/A	N/A
2015/0289254	12/2014	Garcia et al.	N/A	N/A
2015/0289265	12/2014	Gormley et al.	N/A	N/A
2015/0296386	12/2014	Menon et al.	N/A	N/A
2015/0301167	12/2014	Sentelle et al.	N/A	N/A
2015/0319768	12/2014	Abdelmonem et al.	N/A	N/A
2016/0014713	12/2015	Kennedy et al.	N/A	N/A
2016/0037550	12/2015	Barabell et al.	N/A	N/A
2016/0050690	12/2015	Yun et al.	N/A	N/A
2016/0061583	12/2015	Ryu et al.	N/A	N/A
2016/0069994	12/2015	Allen et al.	N/A	N/A
2016/0072597	12/2015	Carbajal	N/A	N/A
2016/0073318	12/2015	Aguirre	N/A	N/A
2016/0080955	12/2015	Carbajal	N/A	N/A
2016/0086621	12/2015	Hearing et al.	N/A	N/A
2016/0095188	12/2015	Verberkt et al.	N/A	N/A
2016/0117853	12/2015	Zhong et al.	N/A	N/A
2016/0124071	12/2015	Baxley et al.	N/A	N/A
2016/0126988	12/2015	Mester et al.	N/A	N/A
2016/0127110	12/2015	McMeekin et al.	N/A	N/A
2016/0127392	12/2015	Baxley et al.	N/A	N/A
2016/0154406	12/2015	Im et al.	N/A	N/A
2016/0161578	12/2015	Weissler et al.	N/A	N/A
2016/0198471	12/2015	Young et al.	N/A	N/A
2016/0219506	12/2015	Pratt et al.	N/A	N/A
2016/0219590	12/2015	Khawer et al.	N/A	N/A

2016/0225240	12/2015	Voddhi et al.	N/A	N/A
2016/0241910	12/2015	Rowe	N/A	N/A
2016/0245921	12/2015	Li et al.	N/A	N/A
2016/0252342	12/2015	Feldmann et al.	N/A	N/A
2016/0323920	12/2015	Carbajal	N/A	N/A
2016/0334527	12/2015	Xu et al.	N/A	N/A
2016/0345135	12/2015	Garcia et al.	N/A	N/A
2016/0364079	12/2015	Qiu et al.	N/A	N/A
2016/0366685	12/2015	Gormley et al.	N/A	N/A
2016/0374088	12/2015	Garcia et al.	N/A	N/A
2017/0024767	12/2016	Johnson, Jr. et al.	N/A	N/A
2017/0025996	12/2016	Cheung et al.	N/A	N/A
2017/0039413	12/2016	Nadler	N/A	N/A
2017/0048838	12/2016	Chrisikos et al.	N/A	N/A
2017/0061690	12/2016	Laughlin et al.	N/A	N/A
2017/0064564	12/2016	Yun et al.	N/A	N/A
2017/0078792	12/2016	Simons	N/A	N/A
2017/0079007	12/2016	Carbajal	N/A	N/A
2017/0094527	12/2016	Shattil et al.	N/A	N/A
2017/0118049	12/2016	Miao et al.	N/A	N/A
2017/0134631	12/2016	Zhao et al.	N/A	N/A
2017/0146462	12/2016	Baker et al.	N/A	N/A
2017/0148332	12/2016	Ziemba et al.	N/A	N/A
2017/0148467	12/2016	Franklin et al.	N/A	N/A
2017/0192089	12/2016	Parker et al.	N/A	N/A
2017/0208495	12/2016	Kleinbeck et al.	N/A	N/A
2017/0234979	12/2016	Mathews et al.	N/A	N/A
2017/0237484	12/2016	Heath et al.	N/A	N/A
2017/0238201	12/2016	Gormley et al.	N/A	N/A
2017/0238203	12/2016	Dzierwa et al.	N/A	N/A
2017/0243138	12/2016	Dzierwa et al.	N/A	N/A
2017/0243139	12/2016	Dzierwa et al.	N/A	N/A
2017/0248677	12/2016	Mahmood et al.	N/A	N/A
2017/0248807	12/2016	Jiang et al.	N/A	N/A
2017/0250766	12/2016	Dzierwa et al.	N/A	N/A
2017/0253330	12/2016	Saigh et al.	N/A	N/A
2017/0261604	12/2016	Van Voorst	N/A	N/A
2017/0261613	12/2016	Van Voorst	N/A	N/A
2017/0261615	12/2016	Ying et al.	N/A	N/A
2017/0274992	12/2016	Chretien	N/A	N/A
2017/0289840	12/2016	Sung et al.	N/A	N/A
2017/0290075	12/2016	Carbajal et al.	N/A	N/A
2017/0311307	12/2016	Negus et al.	N/A	N/A
2017/0358103	12/2016	Shao et al.	N/A	N/A
2017/0366361	12/2016	Afkhami et al.	N/A	N/A
2017/0374572	12/2016	Kleinbeck et al.	N/A	N/A
2017/0374573	12/2016	Kleinbeck et al.	N/A	N/A
2018/0006730	12/2017	Kuo et al.	N/A	N/A
2018/0014217	12/2017	Kleinbeck et al.	N/A	N/A
2018/0024220	12/2017	Massarella et al.	N/A	N/A

2018/0046869	12/2017	Cordell et al.	N/A	N/A
2018/0070362	12/2017	Ryan et al.	N/A	N/A
2018/0074170	12/2017	Ray	N/A	N/A
2018/0074171	12/2017	Ray et al.	N/A	N/A
2018/0081355	12/2017	Magy et al.	N/A	N/A
2018/0083721	12/2017	Wada et al.	N/A	N/A
2018/0124533	12/2017	Messier et al.	N/A	N/A
2018/0129881	12/2017	Seeber et al.	N/A	N/A
2018/0131445	12/2017	Esman et al.	N/A	N/A
2018/0143314	12/2017	Pelletier et al.	N/A	N/A
2018/0149729	12/2017	Grandin et al.	N/A	N/A
2018/0211179	12/2017	Dzierwa	N/A	N/A
2018/0284758	12/2017	Cella et al.	N/A	N/A
2018/0288620	12/2017	Jayawickrama et al.	N/A	N/A
2018/0294901	12/2017	Garcia et al.	N/A	N/A
2018/0313877	12/2017	Brant et al.	N/A	N/A
2018/0313945	12/2017	Parker et al.	N/A	N/A
2018/0324595	12/2017	Shima	N/A	N/A
2018/0329020	12/2017	Hafizovic et al.	N/A	N/A
2018/0331863	12/2017	Carbajal	N/A	N/A
2019/0004518	12/2018	Zhou et al.	N/A	N/A
2019/0011534	12/2018	Trotta et al.	N/A	N/A
2019/0018103	12/2018	Qian et al.	N/A	N/A
2019/0064130	12/2018	Kanazawa et al.	N/A	N/A
2019/0064223	12/2018	Kincaid	N/A	N/A
2019/0072601	12/2018	Dzierwa et al.	N/A	N/A
2019/0074802	12/2018	Geha et al.	N/A	N/A
2019/0077507	12/2018	Ferris et al.	N/A	N/A
2019/0123428	12/2018	Packer et al.	N/A	N/A
2019/0180630	12/2018	Kleinbeck	N/A	N/A
2019/0190553	12/2018	Tsuji et al.	N/A	N/A
2019/0191313	12/2018	Dzierwa et al.	N/A	N/A
2019/0200303	12/2018	Nakahara	N/A	N/A
2019/0208112	12/2018	Kleinbeck	N/A	N/A
2019/0208491	12/2018	Dzierwa et al.	N/A	N/A
2019/0215709	12/2018	Kleinbeck et al.	N/A	N/A
2019/0223139	12/2018	Kleinbeck et al.	N/A	N/A
2019/0230539	12/2018	Dzierwa et al.	N/A	N/A
2019/0230540	12/2018	Carbajal et al.	N/A	N/A
2019/0236266	12/2018	Nashimoto et al.	N/A	N/A
2019/0245722	12/2018	Carbajal	N/A	N/A
2019/0246304	12/2018	Dzierwa et al.	N/A	N/A
2019/0253160	12/2018	Garcia et al.	N/A	N/A
2019/0253905	12/2018	Kleinbeck et al.	N/A	N/A
2019/0260768	12/2018	Mestha et al.	N/A	N/A
2019/0274059	12/2018	Kleinbeck et al.	N/A	N/A
2019/0296910	12/2018	Cheung	N/A	N/A
2019/0302249	12/2018	High et al.	N/A	N/A
2019/0342202	12/2018	Ryan et al.	N/A	N/A
2019/0346571	12/2018	Furumoto	N/A	N/A

2019/0360783	12/2018	Whittaker	N/A	N/A
2019/0364533	12/2018	Kleinbeck et al.	N/A	N/A
2020/0034620	12/2019	Lutterodt	N/A	N/A
2020/0036459	12/2019	Menon	N/A	N/A
2020/0036487	12/2019	Hammond et al.	N/A	N/A
2020/0043346	12/2019	Vacek	N/A	N/A
2020/0059800	12/2019	Menon et al.	N/A	N/A
2020/0066132	12/2019	Kleinbeck	N/A	N/A
2020/0067752	12/2019	DelMarco	N/A	N/A
2020/0068573	12/2019	Drozd et al.	N/A	N/A
2020/0096548	12/2019	Dzierwa et al.	N/A	N/A
2020/0107207	12/2019	Kleinbeck et al.	N/A	N/A
2020/0120266	12/2019	Kleinbeck	N/A	N/A
2020/0128418	12/2019	Dzierwa et al.	N/A	N/A
2020/0137583	12/2019	Economy et al.	N/A	N/A
2020/0142029	12/2019	Brooker et al.	N/A	N/A
2020/0145032	12/2019	Ayala et al.	N/A	N/A
2020/0162890	12/2019	Spencer et al.	N/A	N/A
2020/0167196	12/2019	Smith et al.	N/A	N/A
2020/0169892	12/2019	Dzierwa et al.	N/A	N/A
2020/0184832	12/2019	Kleinbeck	N/A	N/A
2020/0196269	12/2019	Dzierwa et al.	N/A	N/A
2020/0196270	12/2019	Kleinbeck et al.	N/A	N/A
2020/0242603	12/2019	Salkintzis	N/A	N/A
2020/0245167	12/2019	Kleinbeck et al.	N/A	N/A
2020/0260306	12/2019	Kleinbeck et al.	N/A	N/A
2020/0294032	12/2019	Cheng et al.	N/A	N/A
2020/0295855	12/2019	Kleinbeck et al.	N/A	N/A
2020/0382961	12/2019	Shattil et al.	N/A	N/A
2020/0388036	12/2019	Skrede et al.	N/A	N/A
2021/0014696	12/2020	Brookes	N/A	N/A
2021/0067974	12/2020	Guo et al.	N/A	N/A
2021/0082254	12/2020	Givant	N/A	N/A
2021/0084217	12/2020	Kleinbeck	N/A	N/A
2021/0211911	12/2020	Kleinbeck et al.	N/A	N/A
2021/0250795	12/2020	Dzierwa et al.	N/A	N/A
2021/0255356	12/2020	Vu et al.	N/A	N/A
2021/0280039	12/2020	Kleinbeck	N/A	N/A
2021/0281510	12/2020	Brown, Jr. et al.	N/A	N/A
2021/0306022	12/2020	Fernando et al.	N/A	N/A
2021/0360423	12/2020	Dzierwa et al.	N/A	N/A
2021/0360450	12/2020	Kleinbeck et al.	N/A	N/A
2021/0360453	12/2020	Kleinbeck et al.	N/A	N/A
2021/0360454	12/2020	Carbajal et al.	N/A	N/A
2021/0409591	12/2020	Kleinbeck	N/A	N/A
2022/0030541	12/2021	Dzierwa et al.	N/A	N/A
2022/0052770	12/2021	Kleinbeck et al.	N/A	N/A
2022/0128612	12/2021	Dzierwa et al.	N/A	N/A
2022/0131623	12/2021	Garcia et al.	N/A	N/A
2022/0150824	12/2021	Kleinbeck et al.	N/A	N/A

2022/0174525	12/2021	Dzierwa et al.	N/A	N/A
2022/0253407	12/2021	Tran	N/A	N/A
2022/0262228	12/2021	Kleinbeck	N/A	N/A
2022/0262261	12/2021	Kleinbeck	N/A	N/A
2022/0286997	12/2021	Kleinbeck et al.	N/A	N/A
2022/0376921	12/2021	Maria	N/A	N/A
2023/0018133	12/2022	Burnette et al.	N/A	N/A
2023/0045636	12/2022	Dzierwa et al.	N/A	N/A
2023/0087729	12/2022	Goldstein et al.	N/A	N/A
2023/0098387	12/2022	Hafeez et al.	N/A	N/A
2023/0105718	12/2022	Carbajal	N/A	N/A
2023/0114804	12/2022	Kleinbeck	N/A	N/A
2023/0116761	12/2022	Barry et al.	N/A	N/A
2023/0118723	12/2022	Carbajal et al.	N/A	N/A
2023/0123375	12/2022	Dzierwa et al.	N/A	N/A
2023/0126223	12/2022	Kleinbeck et al.	N/A	N/A
2023/0189010	12/2022	Muhammad et al.	N/A	N/A
2023/0206216	12/2022	Lehmann et al.	N/A	N/A
2023/0209378	12/2022	Kleinbeck et al.	N/A	N/A
2023/0232244	12/2022	Dzierwa et al.	N/A	N/A
2023/0252744	12/2022	Miller et al.	N/A	N/A
2023/0254054	12/2022	Garcia et al.	N/A	N/A
2023/0254567	12/2022	Kleinbeck	N/A	N/A
2023/0254702	12/2022	Damnjanovic et al.	N/A	N/A
2023/0275791	12/2022	Carbajal	N/A	N/A
2023/0276280	12/2022	Kleinbeck et al.	N/A	N/A
2023/0300256	12/2022	Zhao et al.	N/A	N/A
2023/0308789	12/2022	Tian et al.	N/A	N/A
2023/0308915	12/2022	Carbajal et al.	N/A	N/A
2023/0326323	12/2022	Kleinbeck	N/A	N/A
2023/0334472	12/2022	Pene	N/A	N/A
2023/0345441	12/2022	Baxley	N/A	N/A
2023/0349962	12/2022	Dzierwa et al.	N/A	N/A
2023/0378645	12/2022	Tran	N/A	N/A
2023/0403564	12/2022	Dzierwa et al.	N/A	N/A
2024/0007204	12/2023	Kleinbeck et al.	N/A	N/A
2024/0023054	12/2023	Dzierwa et al.	N/A	N/A
2024/0029572	12/2023	Kleinbeck	N/A	N/A
2024/0031042	12/2023	Garcia et al.	N/A	N/A
2024/0032084	12/2023	Hellwig	N/A	N/A
2024/0089130	12/2023	Wang et al.	N/A	N/A
2024/0097951	12/2023	Carbajal	N/A	N/A
2024/0103059	12/2023	Dzierwa et al.	N/A	N/A
2024/0114370	12/2023	Kleinbeck et al.	N/A	N/A
2024/0160791	12/2023	He et al.	N/A	N/A
2024/0241163	12/2023	Dzierwa et al.	N/A	N/A
2024/0242589	12/2023	Kleinbeck	N/A	N/A
2024/0244461	12/2023	Kleinbeck et al.	N/A	N/A
2024/0267268	12/2023	Carbajal	N/A	N/A
2024/0267769	12/2023	Dzierwa et al.	N/A	N/A

2024/0275504	12/2023	Kleinbeck et al.	N/A	N/A
2024/0276261	12/2023	Kleinbeck et al.	N/A	N/A
2024/0284214	12/2023	Kleinbeck et al.	N/A	N/A
2024/0292248	12/2023	Kleinbeck et al.	N/A	N/A
2024/0330927	12/2023	Abdelrahman et al.	N/A	N/A
2024/0362631	12/2023	Agrawal et al.	N/A	N/A
2024/0373463	12/2023	Furuichi et al.	N/A	N/A
2024/0378982	12/2023	Kleinbeck	N/A	N/A
2024/0381106	12/2023	Montalvo et al.	N/A	N/A
2024/0386800	12/2023	Kleinbeck et al.	N/A	N/A
2024/0386801	12/2023	Kleinbeck et al.	N/A	N/A
2024/0388937	12/2023	Murias et al.	N/A	N/A
2024/0396648	12/2023	Kleinbeck et al.	N/A	N/A
2024/0430142	12/2023	Carbajal	N/A	N/A
2025/0007629	12/2024	Dzierwa et al.	N/A	N/A
2025/0039711	12/2024	Dzierwa et al.	N/A	N/A
2025/0063405	12/2024	Carbajal et al.	N/A	N/A
2025/0071584	12/2024	Kleinbeck et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
110035440	12/2018	CN	N/A
115482071	12/2021	CN	N/A
117196617	12/2022	CN	N/A
117998368	12/2023	CN	N/A
100248671	12/1999	KR	N/A
20140041618	12/2013	KR	N/A
953557	12/1981	SU	N/A
2012129932	12/2011	WO	N/A
2018184682	12/2017	WO	N/A
2023197982	12/2022	WO	N/A
2023232010	12/2022	WO	N/A

OTHER PUBLICATIONS

“A Hardware Design for Time Delay Estimation of TDOA”; Li et al.; 2013 IEEE International Conference on Signal Processing, Communication and Computing (ICSPCC 2013); Aug. 2013 (Year: 2013). cited by applicant

“A Low-Cost, Near-Real-Time Two-LIAS-Based UWB Emitter Monitoring System”; Wang et al.; IEEE A&E Systems Magazine Nov. 2015 (Year: 2015). cited by applicant

“Joint TDOA and FDOA Estimation: A Conditional Bound and Its Use for Optimally Weighted Localization”; Yeredor et al.; IEEE Transactions on Signal Processing, vol. 59, No. 4, Apr. 2011 (Year: 2011). cited by applicant

“Multipath TDOA and FDOA Estimation Using the EM Algorithm”; Belanger; Apr. 27, 1993; 1993 IEEE International Conference on Acoustics, Speech, and Signal Processing (Year: 1993). cited by applicant

“Noise Figure”, Wikipedia, located at https://en.wikipedia.org/wiki/Noise_figure (Year: 2022). cited by applicant

“Signal Models for TDOA/FDOA Estimation”; Fowler et al.; IEEE Transactions on Aerospace and Electronic Systems vol. 44, No. Oct. 4, 2008 (Year: 2008). cited by applicant

“Specific attenuation model for rain for use in prediction methods”, Recommendation ITU-R p. 838-3 (Year: 2005). cited by applicant

Bluetooth vs Zigbee-difference between Bluetooth and Zigbee (located at <https://www.rfwireless-world.com/Terminology/Bluetooth-vs-zigbee.html>) (Year: 2012). cited by applicant

Boll S.F., Suppression of Acoustic Noise in Speech Using Spectral Subtraction, Apr. 1979, IEEE Transactions on Acoustics, Speech, and Signal Processing, vol. ASSP-27, No. 2, (Year: 1979). cited by applicant

David Eppink and Wolf Kuebler, “TIREM/SEM Handbook”, Mar. 1994, IIT Research Institute, p. 1-6, located at <http://www.dtic.mil/cgi-bin/GetTRDoc?Location=U2&doc=GetTRDoc.pdf&AD=ADA296913>. cited by applicant

English translation of SU-953557-A1 (Year: 2024). cited by applicant

Gabriel Garcia and Daniel Carbajal, U.S. Appl. No. 61/789,758, Provisional Patent Application, filed Mar. 15, 2013 (Specification, Claims, and Drawings). cited by applicant

Gary L. Sugar, System and method for locating wireless devices in an unsynchronized wireless network, U.S. Appl. No. 60/319,737, Provisional Patent Application filed on Nov. 27, 2002, Specification including the claims, abstract, and drawings. cited by applicant

IEEE 100 the Authoritative Dictionary of IEEE Standards Terms. Seventh Edition. Published by Standards Information Network IEEE Press. p. 6 (Year: 2000). cited by applicant

International Search Report and Written Opinion dated Jun. 21, 2018 issued by the International Application Division, Korean Intellectual Property Office as International Searching Authority in connection with International Application No. PCT/US2018/014504 (21 pages). cited by applicant

Mehmet Ali Aygul, Ahmed Naeem, Huseyin Arslan. “Blind Signal Analysis—Wireless Communication Signals”, located at <https://doi.org/10.1002/9781119764441.ch12> (Year: 2021). cited by applicant

Mobile Emitter Geolocation and Tracking Using TDOA and FDOA Measurements; Musicki et al.; IEEE Transactions on Signal Processing, vol. 58, No. 3, Mar. 2010 (Year: 2010). cited by applicant

RF and Digital Signal Processing for Software-Defined Radio, Chapter 4#High-Level Requirements and Link Budget Analysis (Year: 2009). cited by applicant

S. Dörner, S. Cammerer, J. Hoydis and S. t. Brink, “Deep Learning Based Communication Over the Air,” in IEEE Journal of Selected Topics in Signal Processing, vol. 12, No. 1, pp. 132-143, Feb. 2018, doi: 10.1109/JSTSP.2017.2784180. cited by applicant

Steven W. Smith, The Scientist & Engineer's Guide to Digital Signal Processing, 1999, California Technical Publishing, San Diego, California, 2nd Edition, p. 312 (located at http://www.analog.com/media/en/technical-documentation/dsp-book/dsp_book_ch18.pdf) (Year: 1999). cited by applicant

T. J. O'Shea, K. Karra and T. C. Clancy, “Learning to communicate: Channel auto-encoders, domain specific regularizers, and attention,” 2016 IEEE International Symposium on Signal Processing and Information Technology (ISSPIT), Limassol, Cyprus, 2016, pp. 223-228, doi: 10.1109/ISSPIT.2016.7886039. cited by applicant

T. O'Shea and J. Hoydis, “An Introduction to Deep Learning for the Physical Layer,” in IEEE Transactions on Cognitive Communications and Networking, vol. 3, No. 4, pp. 563-575, Dec. 2017, doi: 10.1109/TCCN.2017.2758370. cited by applicant

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Attorney, Agent or Firm: NEO IP

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application relates to and claims priority from the following applications. This application is a continuation of U.S. patent application Ser. No. 18/985,777, filed Dec. 18, 2024, which is a continuation of U.S. patent application Ser. No. 18/789,011, filed Jul. 30, 2024, which is a continuation of U.S. patent application Ser. No. 18/785,936, filed Jul. 26, 2024, which is a continuation-in-part of U.S. patent application Ser. No. 18/775,710, filed Jul. 17, 2024, which is a continuation-in-part of U.S. patent application Ser. No. 18/428,606, filed Jan. 31, 2024, which is a continuation of U.S. patent application Ser. No. 18/374,376, filed Sep. 28, 2023, which is a continuation of U.S. patent application Ser. No. 18/142,904, filed May 3, 2023, which is a continuation of U.S. patent application Ser. No. 17/991,348, filed Nov. 21, 2022, which is a continuation of U.S. patent application Ser. No. 17/735,615, filed May 3, 2022, which is a continuation of U.S. patent application Ser. No. 17/190,048 filed Mar. 2, 2021, which is a continuation of U.S. patent application Ser. No. 16/732,811 filed Jan. 2, 2020, which is a continuation of U.S. patent application Ser. No. 16/275,575 filed Feb. 14, 2019, which claims the benefit of U.S. Provisional Application 62/632,276 filed Feb. 19, 2018. U.S. patent application Ser. No. 16/275,575 also claims priority from and is a continuation-in-part of U.S. patent application Ser. No. 16/274,933 filed Feb. 13, 2019, which is a continuation-in-part of U.S. patent application Ser. No. 16/180,690 filed Nov. 5, 2018, which is a continuation-in-part of U.S. patent application Ser. No. 15/412,982 filed Jan. 23, 2017. U.S. patent application Ser. No. 16/180,690 also claims priority from U.S. Provisional Patent Application No. 62/722,420 filed Aug. 24, 2018. U.S. patent application Ser. No. 16/274,933 also claims the benefit of U.S. Provisional Application 62/632,276 filed Feb. 19, 2018. Each of the above-mentioned applications is incorporated herein by reference in its entirety.

BACKGROUND OF THE INVENTION

1. Field of the Invention

(1) The present invention relates to systems and methods for unmanned vehicle recognition and threat management. More particularly, the systems and methods of the present invention are directed to unmanned vehicle detection, classification and direction finding.

2. Description of the Prior Art

(2) Unmanned Aerial Vehicles (UAVs), commonly known as drones, have become readily available in commercial and retail stores. Detailed schematics for their control systems are available from many manufacturers and the internet along with Software Development Kits (SDKs). Rapid modifications are enabled by advancements in various technologies (e.g., 3D printing). UAVs can be modified to deploy dangerous actions and threaten societal securities. For example, UAVs can be modified to deliver dangerous payloads. It is no longer a question of if, it is now a question of when. Thus, it is imperative that organizations and governments take steps to protect critical assets (e.g., ports, power plants), structures (e.g., buildings, stadiums), and personnel and their citizens.

(3) Exemplary U.S. Patent Documents relevant to the prior art include:

(4) U.S. Pat. No. 9,862,489 for “Method and apparatus for drone detection and disablement” by inventors Lee Weinstein et al., filed Feb. 7, 2016 and issued Jan. 9, 2018, describes a method and apparatus for detection and disablement of an unidentified aerial vehicle (UAV) includes arrays of antenna elements receiving in two modalities (for instance radio frequency (RF) and acoustic modalities, or RF and optical modalities). Signal processing of outputs from multiple antenna arrays locates a potential UAV at specific coordinates within a volume of space under surveillance, and automatically aims video surveillance and a short-range projectile launcher at the UAV, and may automatically fire the projectile launcher to down the UAV.

(5) U.S. Pat. No. 9,858,947 for “Drone detection and classification methods and apparatus” by inventors Brian Hearing et al., filed Nov. 24, 2015 and issued Jan. 2, 2018, describes a system, method, and apparatus for drone detection and classification. An example method includes receiving a sound signal in a microphone and recording, via a sound card, a digital sound sample of

the sound signal, the digital sound sample having a predetermined duration. The method also includes processing, via a processor, the digital sound sample into a feature frequency spectrum. The method further includes applying, via the processor, broad spectrum matching to compare the feature frequency spectrum to at least one drone sound signature stored in a database, the at least one drone sound signature corresponding to a flight characteristic of a drone model. The method moreover includes, conditioned on matching the feature frequency spectrum to one of the drone sound signatures, transmitting, via the processor, an alert.

(6) U.S. Pat. No. 9,767,699 for “System for and method of detecting drones” by inventors John W. Borghese et al., filed May 14, 2015 and issued Sep. 19, 2017, describes an apparatus and method can provide a warning of a drone or unmanned aerial vehicle in the vicinity of an airport. The apparatus can include at least one antenna directionally disposed at an along the approach or departure path and a detector configured to provide a warning of a presence of sense an unmanned aerial or drone. The warning can be provided in response to a radio frequency signal received by the at least one of the antenna being in a frequency band associated with a transmission frequency for the unmanned aerial vehicle or drone or in a frequency band associated with interaction from receive circuitry of the unmanned aerial vehicle or drone.

(7) U.S. Pat. No. 9,715,009 for “Deterrent for unmanned aerial systems” by inventors Dwaine A. Parker et al., filed Dec. 2, 2016 and issued Jul. 25, 2017, describes a system for providing an integrated multi-sensor detection and countermeasure against commercial unmanned aerial systems/vehicles and includes a detecting element, a tracking element, an identification element, and an interdiction element. The detecting element detects an unmanned aerial vehicle in flight in the region of, or approaching, a property, place, event or very important person. The tracking element determines the exact location of the unmanned aerial vehicle. The identification/classification element utilizing data from the other elements generates the identification and threat assessment of the UAS. The interdiction element, based on automated algorithms can either direct the unmanned aerial vehicle away from the property, place, event or very important person in a non-destructive manner, or can disable the unmanned aerial vehicle in a destructive manner. The interdiction process may be over ridden by intervention by a System Operator/HiL.

(8) U.S. Pat. No. 9,529,360 for “System and method for detecting and defeating a drone” by inventors Howard Melamed et al., filed Apr. 22, 2015 and issued Dec. 27, 2016, describes a system for detecting and defeating a drone. The system has a detection antenna array structured and configured to detect the drone and the drone control signal over a 360 degree field relative to the detection antenna array including detecting the directionality of the drone. The system also includes a neutralization system structured and configured in a communicating relation with the detection antenna array. The neutralization system has a transmission antenna structured to transmit an override signal aimed at the direction of the drone, an amplifier configured to boost the gain of the override signal to exceed the signal strength of the drone control signal, and a processing device configured to create and effect the transmission of the override signal. The patent also discloses a method for detecting and defeating a drone.

(9) U.S. Publication No. 2017/0358103 for “Systems and Methods for Tracking Moving Objects” by inventors Michael Shao et al., filed Jun. 9, 2017 and published Dec. 14, 2017, describes systems and methods for tracking moving objects. The publication discloses an object tracking system comprises a processor, a communications interface, and a memory configured to store an object tracking application. The object tracking application configures the processor to receive a sequence of images; estimate and subtract background pixel values from pixels in a sequence of images; compute sets of summed intensity values for different per frame pixel offsets from a sequence of images; identify summed intensity values from a set of summed intensity values exceeding a threshold; cluster identified summed intensity values exceeding the threshold corresponding to single moving objects; and identify a location of at least one moving object in an image based on at

least one summed intensity value cluster.

(10) U.S. Publication No. 2017/0261613 for “Counter drone system” by inventor Brian R. Van Voorst, filed Feb. 27, 2017 and published Sep. 14, 2017, describes a counter drone system that includes a cueing sensor to detect the presence of an object wherein the cueing sensor cues the presence of a target drone, a long range LIDAR system having a sensor pointed in a direction of the target drone to acquire and track at long range the target drone to provide an accurate location of the target drone wherein once a track is acquired, the motion of the target drone is used to maintain the track of the target drone and a threat detector wherein LIDAR data is provided to the threat detector to determine if the target drone is a threat.

(11) U.S. Publication No. 2017/0261604 for “Intercept drone tasked to location of lidar tracked drone” by inventor Brian Van Voorst, filed Feb. 27, 2017 and published Sep. 14, 2017, describes a system that includes a long range LIDAR tracking system to track a target drone and provide detection and tracking information of the target drone; a control system to process the detection and tracking information and provide guidance information to intercept the target drone; and a high powered intercept drone controlled by supervised autonomy, the supervised autonomy provided by processing the detection and tracking information of the target drone and sending guidance information to the intercept drone to direct the intercept drone to the target drone.

(12) U.S. Publication No. 2017/0039413 for “Commercial drone detection” by inventor Gary J. Nadler, filed Aug. 3, 2015 and published Feb. 9, 2017, describes a method of capturing the presence of a drone, including: collecting, using at least one sensor, data associated with an aerial object; analyzing, using a processor, the data to determine at least one characteristic of the aerial object; accessing, in a database, a library of stored characteristics of commercially available drones; determining, based on the analyzing, if the at least one characteristic of the aerial object matches a characteristic of a commercially available drone; and responsive to the determining, generating an indication of a positive match.

(13) U.S. Pat. No. 11,663,992 for Systems and Methods for Detecting, Monitoring, and Mitigating the Presence of Unauthorized Drones by inventors Jordan, et al., Jun. 29, 2020 and published May 23, 2023, is directed to Systems and methods for detecting, monitoring, and mitigating the presence of a drone are provided herein. In one aspect, a system for detecting presence of a drone includes a radio-frequency (RF) receiver. The system can further include a processor and a computer-readable memory in communication with the processor and having stored thereon computer-executable instructions to cause the at least one processor to receive a set of samples from the RF receiver for a time interval, obtain predetermined data of expected communication protocols used between the drone and a controller, and determine whether the RF signal corresponds to one of the expected communication protocols by comparing the samples of the RF signal to the predetermined data and decoding the RF signal. In further aspects the system extracts a unique identifier of the drone based at least partially on the decoded RF signal.

(14) U.S. Pat. No. 11,190,233 for Systems and Methods for Detecting, Monitoring, and Mitigating the Presence of a Drone Using Frequency Hopping by inventors Fang-Hsuan Lo, et al., filed May 28, 2020 and published Nov. 10, 2021, is directed to systems and methods for detecting, monitoring, and mitigating the presence of a drone are provided herein. In one aspect, a system for detecting presence of a one or more drones includes a radio-frequency (RF) receiver configured to receive an RF signal transmitted between a drone and a controller. The system can further include a processor and a computer-readable memory in communication with the processor and having stored thereon computer-executable instructions to cause the at least one processor to receive a set of samples from the RF receiver for a time interval, the set of samples comprising samples of the first RF signal, obtain a parameter model of the first frequency hopping parameters, and fit the parameter model to the set of samples.

(15) U.S. Pat. No. 10,698,076 for Radio Frequency Signal Transmission Detector and Locator by inventors Jones, et al., filed Aug. 2, 2018 and published Jun. 10, 2020, is directed to The present

invention is a system and method for detecting and locating the transmission of radio frequency signals from within a defined geographical area. The system uses statistical confidence limits to detect outliers caused by transmissions in the defined geographical area. The source of the transmission can then be located with triangulation.

(16) U.S. Pat. No. 10,540,905 for Systems, Aircrafts, and Methods for Drone Detection and Collision Avoidance by inventors Bohanan, et al., filed Mar. 28, 2018 and published Jan. 1, 2020, is directed to a system and a method for drone detection and collision avoidance, particularly for use in an aircraft, is provided. The system includes, but is not limited to a sensor, a processor, and an avoidance unit comprising a control unit. The sensor is configured to detect a drone signal in a predetermined space and to transmit the drone signal to the processor. The processor is configured to determine the presence of a drone in the predetermined space based on the drone signal. The processor is configured to transmit a command to the avoidance unit when the processor determines the presence of a drone. The control unit is configured to receive the command and to generate a warning signal in response to receiving the command.

(17) U.S. Pat. No. 9,529,360 for System and Method for Detecting and Defeating a Drone by inventors Melamed, et al., filed Apr. 22, 2015 and published Dec. 7, 2016, is directed to The invention is directed to a system for detecting and defeating a drone. The system has a detection antenna array structured and configured to detect the drone and the drone control signal over a 360 degree field relative to the detection antenna array including detecting the directionality of the drone. The system also includes a neutralization system structured and configured in a communicating relation with the detection antenna array. The neutralization system has a transmission antenna structured to transmit an override signal aimed at the direction of the drone, an amplifier configured to boost the gain of the override signal to exceed the signal strength of the drone control signal, and a processing device configured to create and effect the transmission of the override signal. The invention is also directed to a method for detecting and defeating a drone.

SUMMARY OF THE INVENTION

(18) The present invention provides systems and methods for unmanned vehicle recognition. In one embodiment, a multiplicity of receivers captures RF data and transmits the RF data to at least one node device. The at least one node device comprises a signal processing engine, a detection engine, a classification engine, and a direction finding engine. The at least one node device is configured with an artificial intelligence algorithm. The detection engine and classification engine are trained to detect and classify signals from unmanned vehicles and their controllers based on processed data from the signal processing engine. The direction finding engine is operable to provide lines of bearing for detected unmanned vehicles. A display and control unit is in network communication with the at least one node device for displaying locations and other related data for the detected unmanned vehicles.

(19) In one embodiment, the present invention includes a system for signal identification in a radiofrequency (RF) environment, including at least one node device including a processor and at least one memory in communication with at least one RF receiver, wherein the at least one RF receiver is operable to capture RF data in the RF environment and transmit the RF data to the at least one node device, wherein the at least one node device is operable to average Fast Fourier Transform (FFT) data derived from the RF data into at least one tile, wherein the at least one tile is visually represented as at least one waterfall image, wherein the at least one node device is operable to analyze the at least one waterfall image using machine learning (ML) or at least one convolutional neural network (CNN) to identify at least one signal, at least one signal type, and/or noise to create at least one analyzed waterfall image, and wherein the at least one analyzed waterfall image includes a visual indication of the at least one signal, the at least one signal type, and/or the noise.

(20) In another embodiment, the present invention includes an apparatus for signal identification in a radiofrequency (RF) environment, including a node device including a processor and at least one

memory, wherein the node device is operable to receive RF data from at least one RF receiver, wherein the node device is operable to average Fast Fourier Transform (FFT) data derived from the RF data into at least one tile, wherein the at least one tile is represented as at least one waterfall image, wherein the node device waterfall image is analyzed using machine learning (ML) or at least one convolutional neural network (CNN) to identify at least one signal, at least one signal type, and/or noise to create at least one analyzed waterfall image, and wherein the at least one analyzed waterfall image includes a visual indication of the at least one signal, the at least one signal type, and/or the noise.

(21) In yet another embodiment, the present invention includes a method for signal identification in a radiofrequency (RF) environment, including at least one RF receiver capturing Fast Fourier Transform (FFT) data in the RF environment and transmitting the FFT data to at least one node device, the at least one node device averaging the FFT data derived from the RF data into at least one tile, wherein the at least one tile is represented as at least one waterfall image, the at least one node device analyzing the at least one waterfall image using machine learning (ML) or at least one convolutional neural network (CNN) to identify at least one signal, at least one signal type, and/or noise, and the at least one node device creating at least one analyzed waterfall image based on the identification of the at least one signal, the at least one signal type, and/or the noise, wherein the at least one analyzed waterfall image includes a visual indication of the at least one signal, the at least one signal type, and/or the noise.

(22) These and other aspects of the present invention will become apparent to those skilled in the art after a reading of the following description of the preferred embodiment when considered with the drawings, as they support the claimed invention.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a system for unmanned vehicle recognition according to one embodiment of the present invention.
- (2) FIG. 2 illustrates signal characterization within a spectrum from 700 MHz to 900 MHz.
- (3) FIG. 3 is an illustration of Phantom 4 controller and drone signals.
- (4) FIG. 4 is a human interface display for drone detection according to one embodiment of the present invention.
- (5) FIG. 5 shows a setup for RF data capture in an Anechoic Chamber according to one embodiment of the present invention.
- (6) FIG. 6 illustrates a simulation for fading and channel hopping of an OcuSync drone.
- (7) FIG. 7 is an illustration of an inception based convolutional neural network.
- (8) FIG. 8 illustrates a scenario for an RF environment with just noise.
- (9) FIG. 9 illustrates a scenario for an RF environment with a Phantom 4 controller and drone.
- (10) FIG. 10 illustrates a scenario for an RF environment with two Mavic Pro drones.
- (11) FIG. 11 illustrates a scenario for an RF environment with a Mavic Pro controller only.
- (12) FIG. 12 illustrates a scenario for an RF environment with one Mavic Pro drone only.
- (13) FIG. 13 illustrates a scenario for an RF environment with one Phantom 3 controller only.
- (14) FIG. 14 illustrates a scenario for an RF environment with a Phantom 3 controller and drone.
- (15) FIG. 15 illustrates a scenario for an RF environment with a Mavic Pro controller and drone in wideband mode.
- (16) FIG. 16 illustrates a scenario for an RF environment with a Mavic Pro drone in wideband mode.
- (17) FIG. 17 illustrates a scenario for an RF environment with a Mavic Pro controller and drone.
- (18) FIG. 18 illustrates a scenario for an RF environment with a Mavic Pro controller and a

Phantom 4 controller.

(19) FIG. 19 is an illustration of identifying drones and controllers based on signal edge detection.

(20) FIG. 20 is an illustration with averaged signal amplitudes and signal edges according to one embodiment of the present invention.

(21) FIG. 21 displays a detection range of less than 500 meters based on equipment specification and location.

(22) FIG. 22 displays a training data set containing two separate drone devices operating on the same type of radio protocol according to one embodiment of the present invention.

(23) FIG. 23 displays a training data set containing two separate drone devices operating on different types of radio protocols according to one embodiment of the present invention.

DETAILED DESCRIPTION

(24) The present invention provides systems and methods for unmanned vehicle recognition. The present invention relates to automatic signal detection, temporal feature extraction, geolocation, and edge processing disclosed in U.S. patent application Ser. No. 15/412,982 filed Jan. 23, 2017, U.S. patent application Ser. No. 15/478,916 filed Apr. 4, 2017, U.S. patent application Ser. No. 15/681,521 filed Aug. 21, 2017, U.S. patent application Ser. No. 15/681,540 filed Aug. 21, 2017, and U.S. patent application Ser. No. 15/681,558 filed Aug. 21, 2017, each of which is incorporated herein by reference in its entirety.

(25) In one embodiment, the present invention includes a system for signal identification in a radiofrequency (RF) environment, including at least one node device including a processor and at least one memory in communication with at least one RF receiver, wherein the at least one RF receiver is operable to capture RF data in the RF environment and transmit the RF data to the at least one node device, wherein the at least one node device is operable to average Fast Fourier Transform (FFT) data derived from the RF data into at least one tile, wherein the at least one tile is visually represented as at least one waterfall image, wherein the at least one node device is operable to analyze the at least one waterfall image using machine learning (ML) or at least one convolutional neural network (CNN) to identify at least one signal, at least one signal type, and/or noise to create at least one analyzed waterfall image, and wherein the at least one analyzed waterfall image includes a visual indication of the at least one signal, the at least one signal type, and/or the noise.

(26) In another embodiment, the present invention includes an apparatus for signal identification in a radiofrequency (RF) environment, including a node device including a processor and at least one memory, wherein the node device is operable to receive RF data from at least one RF receiver, wherein the node device is operable to average Fast Fourier Transform (FFT) data derived from the RF data into at least one tile, wherein the at least one tile is represented as at least one waterfall image, wherein the node device waterfall image is analyzed using machine learning (ML) or at least one convolutional neural network (CNN) to identify at least one signal, at least one signal type, and/or noise to create at least one analyzed waterfall image, and wherein the at least one analyzed waterfall image includes a visual indication of the at least one signal, the at least one signal type, and/or the noise.

(27) In yet another embodiment, the present invention includes a method for signal identification in a radiofrequency (RF) environment, including at least one RF receiver capturing Fast Fourier Transform (FFT) data in the RF environment and transmitting the FFT data to at least one node device, the at least one node device averaging the FFT data derived from the RF data into at least one tile, wherein the at least one tile is represented as at least one waterfall image, the at least one node device analyzing the at least one waterfall image using machine learning (ML) or at least one convolutional neural network (CNN) to identify at least one signal, at least one signal type, and/or noise, and the at least one node device creating at least one analyzed waterfall image based on the identification of the at least one signal, the at least one signal type, and/or the noise, wherein the at least one analyzed waterfall image includes a visual indication of the at least one signal, the at least

one signal type, and/or the noise.

(28) Currently, commercial and retail UAVs dominate frequencies including 433 MHz industrial, scientific, and medical radio band (ISM Band) Region 1, 900 MHz ISM Band Region 1,2,3 (varies by country), 2.4 GHz (channels 1-14), 5 GHz (channels 7-165 most predominant), and 3.6 GHz (channels 131-183). Modulation types used by commercial and retail UAVs include Direct Sequence Spread Spectrum (DSSS), Orthogonal Frequency Division Multiplexing (OFDM), Frequency Hopping Spread Spectrum (FHSS), Fataba Advanced Spread Spectrum Technology (FASST).

(29) Many counter UAV systems in the prior art focus on the 2.4 GHz and 5.8 GHz bands utilizing demodulation and decryption of radio frequency (RF) signals to detect and analyze each signal to determine if it is related to a UAV.

(30) The present invention provides systems and methods for unmanned vehicle recognition including detection, classification and direction finding. Unmanned vehicles comprise aerial, terrestrial or water borne unmanned vehicles. The systems and methods for unmanned vehicle recognition are operable to counter threats from the aerial, terrestrial or water borne unmanned vehicles.

(31) In one embodiment, a multiplicity of receivers captures RF data and transmits the RF data to at least one node device. The at least one node device comprises a signal processing engine, a detection engine, a classification engine, and a direction finding engine. The at least one node device is configured with an artificial intelligence algorithm. The detection engine and classification engine are trained to detect and classify signals from unmanned vehicles and their controllers based on processed data from the signal processing engine. The direction finding engine is operable to provide lines of bearing for detected unmanned vehicles. A display and control unit is in network communication with the at least one node device for displaying locations and other related data for the detected unmanned vehicles.

(32) In one embodiment, the present invention provides systems and methods for unmanned vehicle (UV) recognition in a radio frequency (RF) environment. A multiplicity of RF receivers and a displaying device are in network communication with a multiplicity of node devices. The multiplicity of RF receivers is operable to capture the RF data in the RF environment, convert the RF data to fast Fourier transform (FFT) data, and transmit the FFT data to the multiplicity of node devices. The multiplicity of node devices each comprises a signal processing engine, a detection engine, a classification engine, a direction-finding engine, and at least one artificial intelligence (AI) engine. The signal processing engine is operable to average the FFT data into at least one tile. The detection engine is operable to group the FFT data into discrete FFT bins over time, calculate average and standard deviation of power for the discrete FFT bins, and identify at least one signal related to at least one UV and/or corresponding at least one UV controller. The at least one AI engine is operable to generate an output for each of the at least one tile to identify at least one UV and corresponding at least one UV controller with a probability, and calculate an average probability based on the output from each of the at least one tile. The classification engine is operable to classify the at least one UV and/or the at least one UV controller by comparing the at least one signal to classification data stored in a classification library in real time or near real time. The direction-finding engine is operable to calculate a line of bearing for the at least one UV. The displaying device is operable to display a classification of the at least one UV and/or the at least one UV controller and/or the line of bearing of the at least one UV. Each of the at least one tile is visually represented in a waterfall image via a graphical user interface on the displaying device.

(33) FIG. 1 illustrates a system for unmanned vehicle recognition according to one embodiment of the present invention. The system includes a multiplicity of antennas, a receiver and processing unit, and a display and control unit. In one embodiment, there are four multiband omnidirectional antennas. In one embodiment, three multiband omnidirectional antennas are positioned to form an equilateral with 6 meters spacing as illustrated in FIG. 1 as an example. The receiver and

processing unit includes a signal processing engine, a UAV detection engine, a UAV classification engine, a direction finding processing engine, and an internal Global Positioning System (GPS). The receiver and processing unit is operable to receive RF data from the antennas and automatically process the RF data for UAV detection and classification and direction finding. The display and control unit includes a human interface display. In one embodiment, the human interface display is provided by a remote web-based interface. The display and control unit is operable to display lines of bearings for detected UAVs and controllers, classification for detected UAVs and controllers, received signal strength (RSS) values, and operating frequencies. In one embodiment, the display and control unit is SigBase 4000 as shown in FIG. 1. In another embodiment, any computer, laptop, or tablet configured with the human interface display of the present invention is operable to function as a display and control unit. In one embodiment, the receiver and processing unit is a node device, and there are multiple node devices communicating with each other and forming a group of networked nodes for UAV detection, classification, and direction finding.

(34) The present invention provides a more efficient methodology for UAV detection and identification, which takes advantage of Fast Fourier Transform (FFT) over a short period of time and its derivation. RF data received from antennas are directly converted to FFT data with finer granularity. This allows rapid identification of protocols used by high threat drones without demodulation, and the identification is probability based. An analytics engine is operable to perform near real-time analysis and characterize signals within the spectrum under observation. FIG. 2 illustrates signal characterization within a spectrum from 700 MHz to 900 MHz. Temporal feature extraction is applied for signal characterization, which is described in U.S. patent application Ser. No. 15/412,982 filed Jan. 23, 2017, U.S. patent application Ser. No. 15/681,521 filed Aug. 21, 2017, U.S. patent application Ser. No. 15/681,540 filed Aug. 21, 2017, U.S. patent application Ser. No. 15/681,558 filed Aug. 21, 2017, each of which is incorporated herein by reference in its entirety.

(35) Advantageously, multiple receivers in the present invention work together to ingest spectral activities across large blocks of spectrum. The multiple receivers have an instantaneous bandwidth from 40 MHz to 500 MHz. In one embodiment, the multiple receivers are configurable in 40 MHz and 125 MHz segment building blocks. Input data are converted directly to FFT data and fed into process engines, which significantly decreases latency. The process engines are designed for rapid identification of signals of interest (SOI). When an SOI is detected, a direction finding process is initiated autonomously. In one embodiment, the direction finding process is configurable by an operator.

(36) There are multiple types of communications links utilized for command and control of an unmanned vehicle. Although several cost-effective radio communication (RC) protocols are gaining global popularity, WI-FI is still the most popular protocol for command and control of UAVs and camera systems. A remote controller of a UAV acts as a WI-FI access point and the UAV acts as a client. There are several limiting factors for WI-FI-based UAVs. For example, the operational range of a WI-FI-based UAV is typically limited to 150 feet (46 m) indoor and 300 feet (92 m) outdoor. There is significant latency for control and video behaviors. Interference by other WI-FI devices affects operational continuity of the WI-FI-based UAVs.

(37) Demand in the UAV user community has made more professional-level protocols available in the commercial and retail markets. By way of example but not limitation, two common RC protocols used for UAVs are Lightbridge and OcuSync. Enhancements in drone technology inevitably increases the capability of drones for use in industrial espionage and as weapons for nefarious activities.

(38) Lightbridge is developed for long range and reliable communication. Communication is available within a range up to 5 km. Lightbridge supports 8 selectable channels, and the selection can be manual or automatic. Drones with Lightbridge protocol also have the ability to assess

interference and move to alternate channels for greater quality.

(39) OcuSync is developed based on the LightBridge protocol. OcuSync uses effective digital compression and other improvements, which decreases knowledge required to operate. OcuSync provides reliable HD and UHD video, and OcuSync-based drones can be operated in areas with greater dynamic interference. OcuSync improves command and control efficiencies and reduces latency. With OcuSync, video communications are improved substantially, operational range is increased, command and control recovery are enhanced when interference occurs.

(40) The systems and methods of the present invention for unmanned vehicle recognition are operable to detect and classify UAVs at a distance, provide directions of the UAVs, and take defensive measures to mitigate risks. The detection and classification are fast, which provides more time to react and respond to threats. Exact detection range is based upon selection of antenna systems, topology, morphology, and client criteria. Classification of the detected UAVs provides knowledge of the UAVs and defines effective actions and capabilities for countering UAV threats. In one embodiment, the direction information of the UAVs provides orientation within the environment based on the location of the UAV detector.

(41) In one embodiment, the systems and methods of the present invention provides unmanned vehicle recognition solution targeting radio controlled and WI-FI-based drones. The overall system is capable of surveying the spectrum from 20 MHz to 6 GHz, not just the common 2.4 GHz and 5.8 GHz areas as in the prior art. In one embodiment, the systems and methods of the present invention are applied to address 2 major categories: RC-based UAV systems and WI-FI-based UAV systems. In one embodiment, UAV systems utilize RC protocols comprising LightBridge and OcuSync. In another embodiment, UAV systems are WI-FI based, for example but not for limitation, 3DR Solo and Parrot SkyController. The systems and methods of the present invention are operable to detect UAVs and their controllers by protocol.

(42) The systems and methods of the present invention maintain a state-of-the-art learning system and library for classifying detected signals by manufacturer and controller type. The state-of-the-art learning system and library are updated as new protocols emerge.

(43) In one embodiment, classification by protocol chipset is utilized to provide valuable intelligence and knowledge for risk mitigation and threat defense. The valuable intelligence and knowledge include effective operational range, supported peripherals (e.g., external or internal camera, barometers, GPS and dead reckoning capabilities), integrated obstacle avoidance systems, and interference mitigation techniques.

(44) The state-of-the-art learning system of the present invention is highly accurate and capable of assessing detected UAV signals and/or controller signals for classification in less than a few seconds with a high confidence level. The state-of-the-art learning system is operable to discriminate changes in the environment for non-drone signals as well as drone signals. FIG. 3 is an illustration of Phantom 4 controller and drone signals. A human interface is operable to display the classification results. FIG. 4 illustrates a human interface display for drone detection according to one embodiment of the present invention.

(45) It is difficult to recognize commercial and retail drones with the naked eye over 100 meters. It is critical to obtain a vector to the target for situational awareness and defense execution. The systems and methods of the present invention provides lines of bearing for direction finding for multiple UAVs flying simultaneously. Each line of bearing is color coded for display. Angles, along with frequencies utilized for uplink and downlink, are also displayed on the human interface.

(46) Once a UAV is detected and classified, an alert is posted to a counter UAV system operator (e.g., a network operation center, an individual operator) including azimuth of the UAV and other information. The alert is transmitted via email, short message service (SMS) or third-party system integration. The counter UAV system is operable to engage an intercession transmission, which will disrupt the communication between the UAV and its controller. When the communication between the UAV and its controller is intercepted, the UAV will invoke certain safety protocols, such as

reduce height and hover, land, or return to the launch point. The counter UAV system may have certain restrictions based on country and classification of the UAV.

(47) In one embodiment, the systems and methods of the present invention are operable to update the UAV library with emerging protocols for classification purposes, and refine the learning engine for wideband spectrum analysis for other potential UAV signatures, emerging protocols and technologies. In other words, the systems and methods of the present invention are adaptable to any new and emerging protocols and technologies developed for unmanned vehicles. In one embodiment, multiple node devices in the present invention are deployed to operate as a group of networked nodes. In one embodiment, the group of networked nodes are operable to estimate geographical locations for unmanned vehicles. In one embodiment, two node devices are operable to provide a single line of bearing and approximate a geographical location of a detected drone or controller. The more node devices there are in the group of network nodes, the more lines of bearing are operable to be provided, and the more accurate the geographical location is estimated for the detected drone or controller. In one embodiment, the geolocation function provides altitude and distance of a targeted drone.

(48) In one embodiment, the counter UAV system in the present invention is operable to alert when unexpected signal characteristics are detected in 2.4 GHz and 5.8 GHz areas and classify the unexpected signal characteristics as potential UAV activities. In another embodiment, the counter UAV system in the present invention is operable to alert when unexpected signal characteristics are detected anywhere from 20 MHz to 6 GHz and classify the unexpected signal characteristics as potential UAV activities. In another embodiment, the counter UAV system in the present invention is operable to classify the unexpected signal characteristics as potential UAV activities when unexpected signal characteristics are detected anywhere from 40 MHz to 6 GHz. The automatic signal detection engine and analytics engine are enhanced in the counter UAV system to recognize potential UAV activities across a great portion of the spectrum. In one embodiment, any blocks of spectrum from 40 MHz to 6 GHz are operable to be selected for UAV recognition.

(49) In one embodiment, vector-based information including inclinations, declinations, topology deviations, and user configurable Northing map orientation is added to the WGS84 mapping system for direction finding and location estimation. In one embodiment, earth-centered earth-fixed vector analysis is provided for multi-node systems to estimate UAV locations, derive UAV velocities from position changes over time, and determine UAV trajectory vectors in fixed nodal processing. In one embodiment, a group of networked node devices are operable to continually provide lines of bearing over time, approximate geographical locations of a detected unmanned vehicle on or above the earth, and track the movement of the detected unmanned vehicle from one estimated location to another. In one embodiment, the group of networked node devices are operable to determine velocities of the detected unmanned vehicle based on estimated locations and travel time. In one embodiment, the group of networked node devices are operable to estimate a trajectory of the detected unmanned vehicle based on the estimated geographical locations over time. In one embodiment, the group of networked node devices are operable to estimate accelerations and decelerations of the unmanned vehicle based on the velocities of the unmanned vehicles over time.

(50) In one embodiment, the systems and methods of the present invention are operable for UAV detection and direction finding for different modulation schemes including but not limited to DSSS, OFDM, FHSS, FASST, etc. In one embodiment, the counter UAV system in the present invention is configured with cameras for motion detection. The cameras have both day and night vision.

(51) In one embodiment, systems and methods of the present invention provides training for unmanned vehicle recognition. RF data is captured for a Phantom 3 drone and its controller and a Phantom 4 drone and its controller, both of which use Lightbridge protocol. RF data is also captured for a Mavic Pro drone and its controller, which uses OcuSync protocol. The RF data is recorded at different channels, different RF bandwidths, and different video quality settings inside

and outside an Anechoic Chamber. FIG. 5 shows a setup for RF data capture in an Anechoic Chamber according to one embodiment of the present invention. The recordings are overlaid on the RF environment, and fading and channel hopping are simulated. FIG. 6 illustrates a simulation for fading and channel hopping of an OcuSync drone.

(52) In one embodiment, the recorded RF data is used to train and calibrate an inception based convolutional neural network comprised in a drone detection system. FIG. 7 is an illustration of an inception based convolutional neural network. U.S. Patent Publication No. 2018/0137406 titled "Efficient Convolutional Neural Networks and Techniques to Reduce Associated Computational Costs" is incorporated herein by reference in its entirety. The inception based convolutional neural network generates probabilities that drones or their controllers are detected. The detection probabilities are updated multiple times per second.

(53) The trained inception based convolutional neural network is operable to identify Lightbridge 1 controller and drone, Lightbridge 2 controller and drone, and OcuSync controller and drone. The trained inception based convolutional neural network is operable to identify Lightbridge and Ocusync controllers and drones at the same time. In one embodiment, the drone detection system comprising the trained inception based convolutional neural network is operable to search an instantaneous bandwidth of 147.2 MHz.

(54) In one embodiment, the drone detection system of the present invention includes an artificial intelligence (AI) algorithm running on a single board computer (e.g., Nvidia Jetson TX2) with an execution time less than 10 ms. The drone detection system is operable to separate Phantom 3 and Phantom 4 controllers. Waveforms for Phantom 3 and Phantom 4 controllers are sufficiently different to assign separate probabilities.

(55) The Artificial Intelligence (AI) algorithm is used to enhance performance for RF data analytics. The RF data analytics process based on the AI algorithm is visualized. The RF waterfalls of several drone scenarios are presented in FIGS. 8-18. FIG. 8 illustrates a scenario for an RF environment with just noise. FIG. 9 illustrates a scenario for an RF environment with a Phantom 4 controller and drone. FIG. 10 illustrates a scenario for an RF environment with two Mavic Pro drones. FIG. 11 illustrates a scenario for an RF environment with a Mavic Pro controller only. FIG. 12 illustrates a scenario for an RF environment with one Mavic Pro drone only. FIG. 13 illustrates a scenario for an RF environment with one Phantom 3 controller only. FIG. 14 illustrates a scenario for an RF environment with a Phantom 3 controller and drone. FIG. 15 illustrates a scenario for an RF environment with a Mavic Pro controller and drone in wideband mode. FIG. 16 illustrates a scenario for an RF environment with a Mavic Pro drone in wideband mode. FIG. 17 illustrates a scenario for an RF environment with a Mavic Pro controller and drone. FIG. 18 illustrates a scenario for an RF environment with a Mavic Pro controller and a Phantom 4 controller.

(56) Each scenario is illustrated with 6 waterfall images. Each waterfall represents ~80 ms of time and 125 MHz of bandwidth. The top left image is the waterfall before an AI processing. The other five images are waterfalls after the AI processing. For each signal type, the areas of the waterfall that are likely for the RF signal type are highlighted. Areas that are not for the signal type are grayed out. The overall probability that a signal exists in the image is printed in the title of each waterfall image. In one embodiment, the AI algorithm is securely integrated with a state engine and a detection process of the present invention. In one embodiment, AI processing or processing of the waterfall includes a comparison of the waterfall image to a database of waterfall images of the RF environment or similar RF environments for which signals or noise were identified. The comparison of the waterfall image to the database of waterfall images is operable to be performed in real time or near real time. In one embodiment, the database of waterfall images is operable to be updated in real time or near real time with the waterfall image created based on the at least one tile and associated information with the waterfall image, including but not limited to the signal identified in the waterfall image, the signal type(s) identified in the waterfall image, and/or noise identified in the waterfall image.

(57) The comparison includes the use of machine learning (ML) or convolutional neural networks (CNN) in one embodiment. In other embodiments, the node or system is operable to utilize a plurality of learning techniques for analyzing waterfall images including, but not limited to, artificial intelligence (AI), deep learning (DL), artificial neural networks (ANNs), support vector machines (SVMs), Markov decision process (MDP), and/or natural language processing (NLP). The node or system is operable to use any of the aforementioned learning techniques alone or in combination. Further, the node or system is operable to utilize predictive analytics techniques including, but not limited to, machine learning (ML), artificial intelligence (AI), neural networks (NNs) (e.g., long short term memory (LSTM) neural networks), deep learning, historical data, and/or data mining to make future predictions and/or models. The node or system is preferably operable to recommend and/or perform actions based on historical data, external data sources, ML, AI, NNs, and/or other learning techniques. The node or system is operable to utilize predictive modeling and/or optimization algorithms including, but not limited to, heuristic algorithms, particle swarm optimization, genetic algorithms, technical analysis descriptors, combinatorial algorithms, quantum optimization algorithms, iterative methods, deep learning techniques, and/or feature selection techniques.

(58) In one embodiment, a method for drone detection and classification comprises applying FFT function to RF data, converting FFT data into logarithmic scale in magnitude, averaging converted FFT into 256 by 256 array representing 125 MHz of bandwidth and 80 ms of time as a base tile, applying normalization function to the base tile, applying a series of convolutional and pooling layers, applying modified You Only Look Once (YOLO) algorithm for detection, grouping bounding boxes displayed in the waterfall images (e.g., waterfall plots in FIGS. 8-18), classifying signals based on the shape of detection output, verifying results with a second level recurrent neural network (RNN) based pattern estimator.

(59) In one embodiment, a method for training comprises recording clean RF signals, shifting RF signals in frequency randomly, creating truth data for YOLO output, adding a simulated channel to the RF signals, recording typical RF backgrounds, applying FFT function to RF data, converting FFT data into logarithmic scale in magnitude, averaging converted FFT into 256 by 256 array representing 125 MHz of bandwidth and 80 ms of time as a base tile, applying normalization function to the base tile, applying a series of convolutional and pooling layers, applying modified You Only Look Once (YOLO) algorithm for detection, grouping bounding boxes displayed in the waterfall images (e.g., waterfall plots in FIGS. 8-18), applying a sigmoid cross entropy function, and applying an Adaptive Moment Estimation (Adam) based back propagation algorithm.

(60) In one embodiment, a drone detection engine is operable to convert FFT flows from a radio to a tile. For each channel, the drone detection engine is operable to standardize the FFT output from the radio at a defined resolution bandwidth, and group high resolution FFT data into distinct bins overtime. The drone detection engine is further operable to calculate average and standard deviation of power for discrete FFT bins, assign a power value to each channel within the tile. Each scan or single stare at the radio is a time slice, and multiple time slices with power and channel assignment create a tile. Tiles from different frequency spans and center frequencies are identified as a tile group by a tile group number. Receivers in the drone detection system are operable to be re-tuned to different frequencies and spans. In one embodiment, the drone detection system comprises multiple receivers to generate tiles and tile groups.

(61) In one embodiment, a tile is sent to a YOLO AI Engine. Outputs of a decision tree in the YOLO AI engine are used to detect multiple drones and their controllers. Drones of the same type of radio protocol are operable to be identified within the tile. Controllers of the same type of radio protocol are operable to be identified within the tile. Drones of different radio protocols are also operable to be identified within the tile. Controllers of different radio protocols are also operable to be identified within the tile. FIG. 19 is an illustration of identifying drones and controllers based on signal edge detection.

(62) In one embodiment, a plurality of tiles is sent to the YOLO AI engine. In one embodiment, a tile group is sent to the YOLO AI engine. The YOLO AI engine generates an output for each tile to identify drones and their controllers with a probability. An average probability is calculated based on outputs for multiple tiles in the tile group. For each tile group, the YOLO AI engine computes outputs for several tiles per second.

(63) In one embodiment, a state engine controls the flows of tiles and tile groups into one or more AI engines. AI engines do not use frequency values for analytics. Thus, the one or more AI engines are operable for any frequency and frequency span that a drone radio supports. The state engine further correlates output of the one or more AI engines to appropriate tiles and tile groups.

(64) The systems and methods of the present invention are operable for direction finding of drones and their controllers. Outputs from the AI engine are denoted with time basis for the drones and their controllers.

(65) Drones typically maintain the same frequency unless their firmware detects interference. Then the drones may negotiate a change with their controllers. This does not create an issue for detection as long as the new frequency and span is monitored by the systems and methods of the present invention. Drone controllers typically use a frequency hopping spread spectrum (FHSS) or other Frequency hopping system (e.g., Gaussian frequency shift keying (GFSK)).

(66) In one embodiment, the systems and method of the present invention are operable to approximate a start time of a line of bearing for a direction finding (DF) system. The time intervals are either known or estimated based upon the behavior monitored by the AI engine and state engine. This allows the time slice and frequency of each individual drone and/or controller to be passed to the DF system. In one embodiment, three or four receivers are coordinated to collect information in appropriate frequency segments, wherein the frequency segments are similar to tiles described earlier. FIG. 20 is an illustration with averaged signal amplitudes and signal edges according to one embodiment of the present invention.

(67) The AI engine examines the segments to determine if a drone or controller exists. An azimuth of the drone or controller in an Earth-Centered Earth-Fixed coordinate system is determined based on other information collected from the three or four receivers using time difference of arrival (TDOA), angle of arrival (AOA), power correlative, or interferometry techniques.

(68) Distance capability of UAV detection and classification system depends on hardware configuration, environment morphology and restrictions based on country and classification of the counter UAV operator. In one embodiment, the systems and methods for unmanned vehicle recognition are operable to detect unmanned vehicles within 3-4 kilometers. FIG. 21 displays a detection range of less than 500 meters based on equipment specification and location.

(69) In one embodiment, the systems and methods of the present invention are operable to utilize training data. In one embodiment training data contains at least one tile displaying at least one waterfall image for one or more drones. These images are numbered based on the time slices and color coded to keep track of each drone. The numbering and color coding of each drone is fed into the ML drone frequency hopping algorithm. The color coding and numbering of each drone signal allows for easy training of the image analysis algorithm to quickly pick up on trends in drone frequency hopping which allows for quick training of the algorithm to account for multiple drones and/or frequency hopping types. In one embodiment, one or more drones contain the same and/or different types of radio protocols within at least one tile. In one embodiment the training data is further utilizable by one or more AI engines operable for any frequency and frequency span supported by drone radio. In one embodiment, the AI engine is operable to use machine learning (ML), artificial intelligence (AI), You Only Look Once Artificial Intelligence (YOLO AI), deep learning (DL), neural networks (NNs), artificial neural networks (ANNs), convolutional neural networks (CNNs), support vector machines (SVMs), Markov decision process (MDP), natural language processing (NLP), control theory, and/or statistical learning techniques. FIG. 22 displays a tile containing drones of the same type of radio protocol. FIG. 23 displays training data of a tile

containing drones of different types of radio protocols.

(70) In one embodiment the AI and/or ML algorithm is operable to classify the drone and/or signal modulation type based on the waterfall image created by the signal. The training data and/or waterfall images are operable to be sets of known modulation schemes for drones for the ML algorithm to provide an image comparison to the signal in the RF environment. The known frequency hopping schemes are operable to be learned by the algorithm and then identified by pattern recognition when a signal in the environment is analyzed and put into a waterfall graphical representation. The image comparison ML or AI algorithm is operable to detect the modulation type patterns of the signal.

(71) In one embodiment, the training data is operable to be utilized for “No Drone Zone” identification and avoidance. The Federal Aviation Administration (FAA) denotes areas where drones are unable to operate as “No Drone Zones”, with each zone containing specific operating restrictions. To report “No Drone Zones” to recreational flyers, the FAA released B4UFLY—a service that allows recreational flyers to determine where drones are operable and not operable.

(72) In one embodiment, the training data is operable to be fed to the ML Drone Frequency Hopping algorithm to allow for identification and avoidance of “No Drone Zones”. In a further embodiment, the ML Drone Frequency Hopping algorithm is operable to cross reference detected signals with B4UFLY to determine if the signal is in a restricted area.

(73) In one embodiment, the training data includes at least a tile or group of tiles representing a drone that is operable to exist in a “No Drone Zone”. In one embodiment, the system utilizes the ML or AI Drone Frequency Hopping algorithm to confirm detection of a drone before cross referencing the signal with B4UFLY, the geolocation engine, and other sensors to determine if the drone has breached a “No Drone Zone”. In one embodiment, other sensors include visual systems such as LIDAR, cameras and radar. The system is operable to interface with the multiple data sources, determine if the information from the multiple data sources is consistent or in agreement, and create a report identifying and determining if the drone has breached a “No Drone Zone”. If the sensor information is not in agreement, the system is operable to discard the data and create a report identifying the faulty data.

(74) In one embodiment, the present invention is operable to utilize training data to detect friendly drones (F-Drone) and nefarious drones (N-Drone). In one embodiment an F-Drone is identified by a specific radio protocol and/or a N-Drone is identified by not fitting into the desired radio protocol. In another embodiment, an N-Drone is identified by a desired radio protocol and/or a F-Drone is identified by not identifying with a targeted radio protocol. In a further embodiment one or more AI engines is operable to detect F-Drones and/or N-Drones utilizing training data. Training data is further utilizable by one or more AI engines that are operable for any frequency and frequency span supported by drone radio. In one embodiment, training data is operable to be utilized by one or more AI engines to define a “No Drone Zone”. In a further embodiment, the AI engine is operable to detect a N-Drones by identifying a signal within the “No Drone Zone”. In one embodiment training data is cross referenced with B4UFLY, the geolocation engine, and other sensors to embody parameters, including but not limited to altitude, to detect friendly and/or nefarious drones. In one embodiment, a report is created and sent identifying and detecting F-Drones and/or N-Drones.

(75) Certain modifications and improvements will occur to those skilled in the art upon a reading of the foregoing description. The above-mentioned examples are provided to serve the purpose of clarifying the aspects of the invention and it will be apparent to one skilled in the art that they do not serve to limit the scope of the invention. All modifications and improvements have been deleted herein for the sake of conciseness and readability but are properly within the scope of the present invention.

Claims

1. A system for signal identification in a radiofrequency (RF) environment, comprising: at least one node device including a processor and memory in communication with at least one RF receiver; wherein the at least one node device is in communication with at least one artificial intelligence (AI) engine; wherein the at least one RF receiver is operable to capture RF data in the RF environment and transmit the RF data to the at least one node device; wherein the at least one node device is operable to average Fast Fourier Transform (FFT) data derived from the RF data into at least one tile; wherein the at least one AI engine is operable to generate an output for the at least one tile to identify and correlate at least one signal with a probability; wherein the at least one tile is represented as at least one waterfall image; and wherein location information for the at least one signal is displayed on at least one user interface as at least one line of bearing.
2. The system of claim 1, wherein the at least one AI engine is operable to calculate an average probability based on the output.
3. The system of claim 1, wherein the at least one AI engine is trained using at least one training data set.
4. The system of claim 1, wherein a signal modulation type of the at least one signal includes direct sequence spread spectrum (DSSS), orthogonal frequency division multiplexing (OFDM), frequency hopping spread spectrum (FHSS), and/or Futaba advanced spread spectrum technology (FASST).
5. The system of claim 1, wherein the at least one signal includes a drone signal.
6. The system of claim 1, wherein the at least one node device is operable to transmit an alert and/or report related to a comparison of the location information of the at least one signal against at least one location of interest.
7. The system of claim 1, wherein the at least one line of bearing is calculated by a direction-finding engine.
8. The system of claim 1, wherein the at least one signal is highlighted on the at least one waterfall image.
9. The system of claim 1, wherein the at least one RF receiver is operable to transform the RF data to the FFT data.
10. An apparatus for signal identification in a radiofrequency (RF) environment, comprising: a node device including a processor and memory; wherein the node device is in communication with at least one artificial intelligence (AI) engine; wherein the node device is operable to receive RF data from at least one RF receiver; wherein the node device is operable to average Fast Fourier Transform (FFT) data derived from the RF data into at least one tile; wherein the at least one AI engine is operable to generate an output for the at least one tile to identify and correspond at least one signal with a probability; wherein the RF data includes location information of the at least one signal; wherein the at least one tile is represented as at least one waterfall image; and wherein the location information for the at least one signal is displayed on at least one user interface as at least one color coded line of bearing.
11. The apparatus of claim 10, wherein the location information includes an airport no-fly-zone and/or at least one geofence.
12. The apparatus of claim 10, wherein the at least one AI engine is operable to calculate an average probability based on the output.
13. The apparatus of claim 10, wherein the at least one node device is operable to transmit an alert and/or report related to a comparison of the location information of the at least one signal against at least one location of interest.
14. The apparatus of claim 10, wherein the at least one AI engine is operable to be trained using at least one training data set.

15. The apparatus of claim 10, wherein the at least one color coded line of bearing is calculated by a direction-finding engine.
16. A method for signal analysis in a radiofrequency (RF) environment, comprising: capturing RF data in the RF environment by at least one RF receiver, converting the RF data to Fast Fourier Transform (FFT) data by the at least one RF receiver and transmitting the FFT data to at least one node device; averaging the FFT data by the at least one node device into at least one tile and representing the at least one tile as at least one waterfall image; generating an output for the at least one tile by at least one artificial intelligence (AI) engine in communication with the at least one node device to identify and correlate at least one signal with a probability; calculating an average probability based on the output of the at least one tile; classifying at least one signal emitting device based by comparing the at least one signal emitted from the at least one signal emitting device to classification data stored in a classification library; and wherein location information for the at least one signal is displayed on at least one user interface as at least one line of bearing.
17. The method of claim 16, wherein the location information includes an airport no-fly-zone and/or at least one geofence.
18. The method of claim 16, further comprising the at least one node device transmitting an alert and/or report related to a comparison of the location information for the at least one signal against at least one location of interest.
19. The method of claim 16, wherein the at least one AI engine is trained using at least one training data set.
20. The method of claim 16, wherein the at least one signal characteristic includes a modulation type of the at least one signal that includes direct sequence spread spectrum (DSSS), orthogonal frequency division multiplexing (OFDM), frequency hopping spread spectrum (FHSS), and/or Futaba advanced spread spectrum technology (FASST).
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